

BFS Traversal

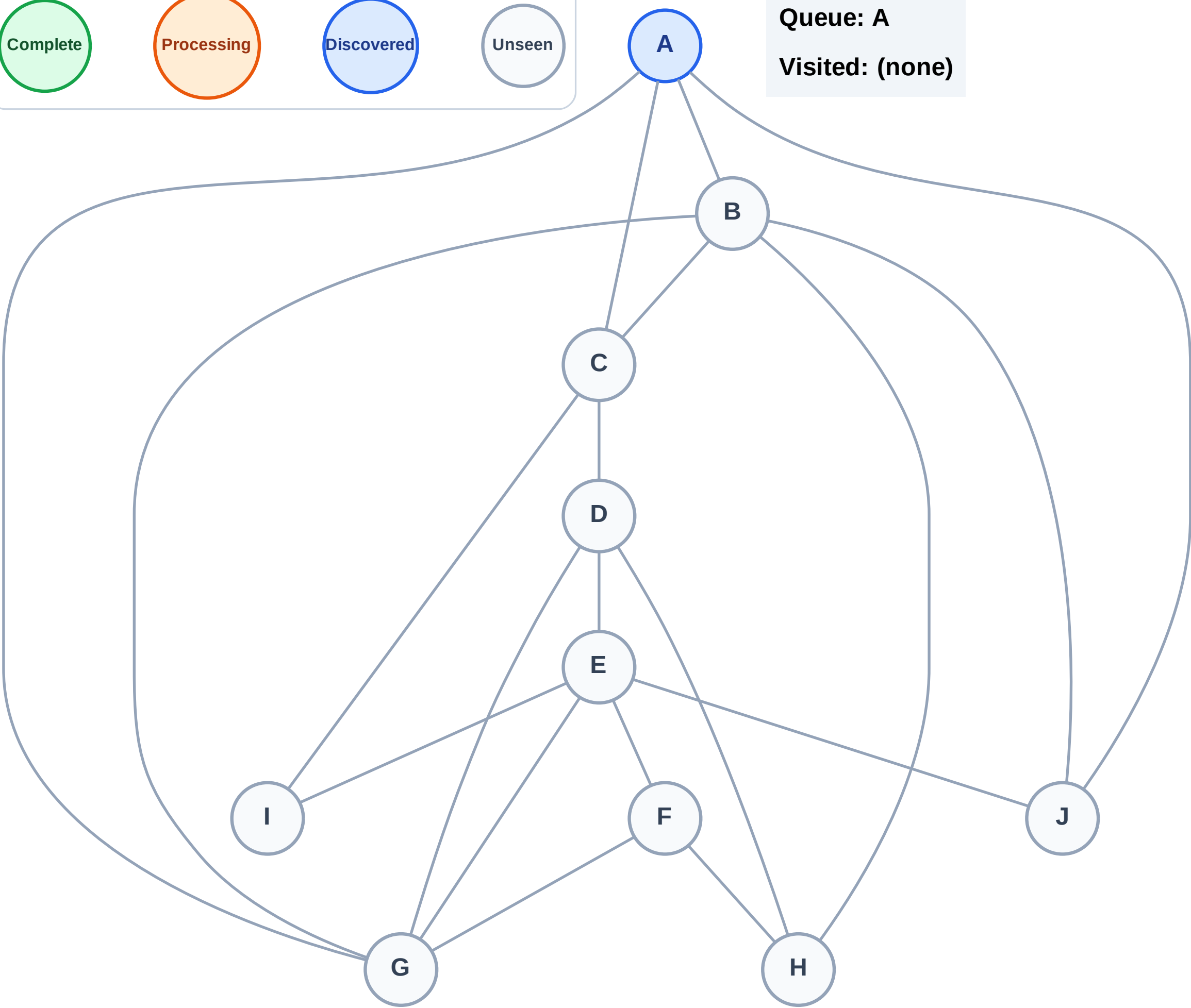
Step 1: Start at A

Legend



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

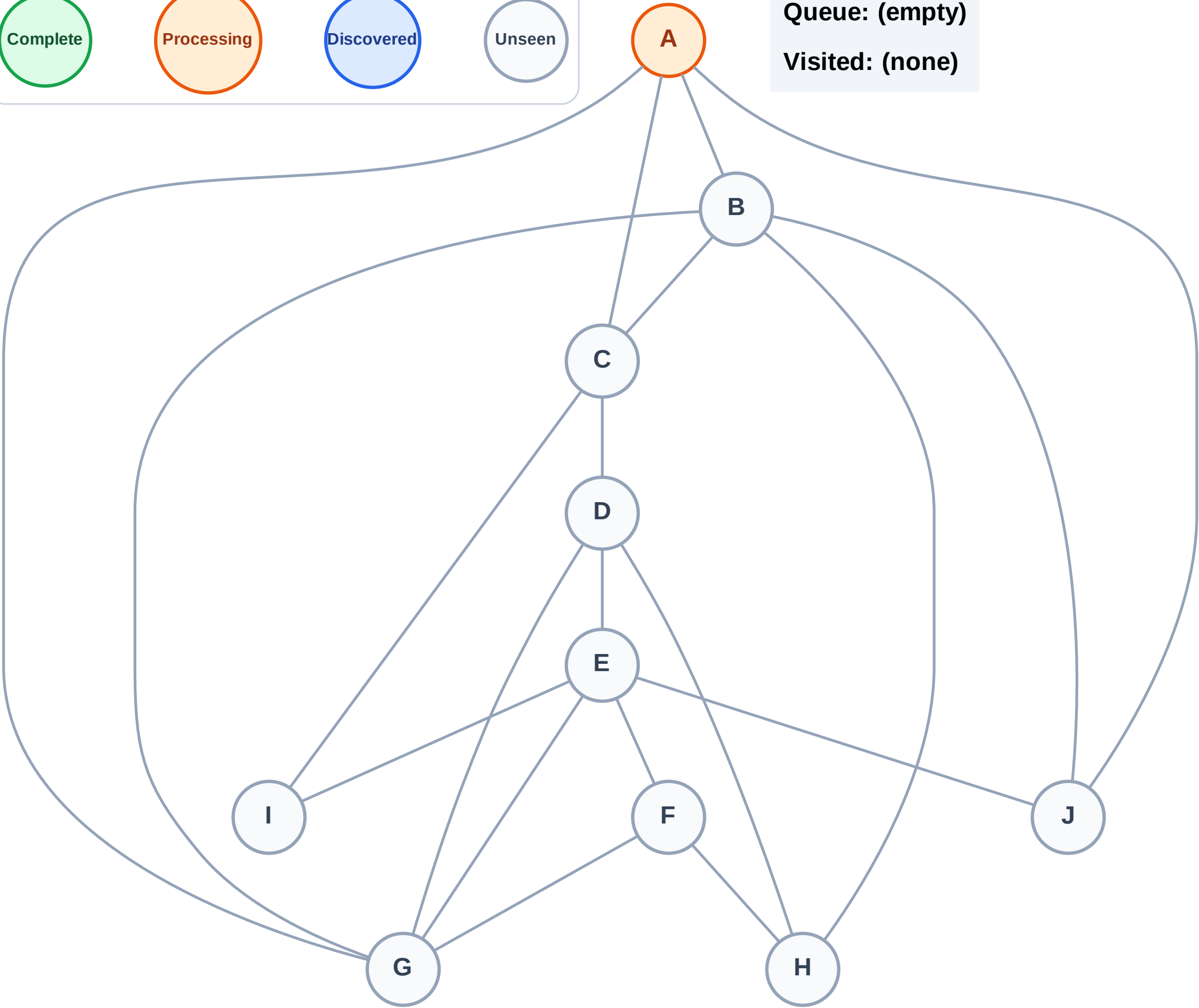
Complete

Processing

Discovered

Unseen

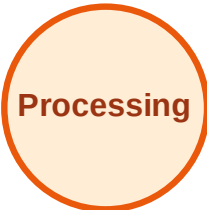
Queue: (empty)
Visited: (none)



BFS Traversal

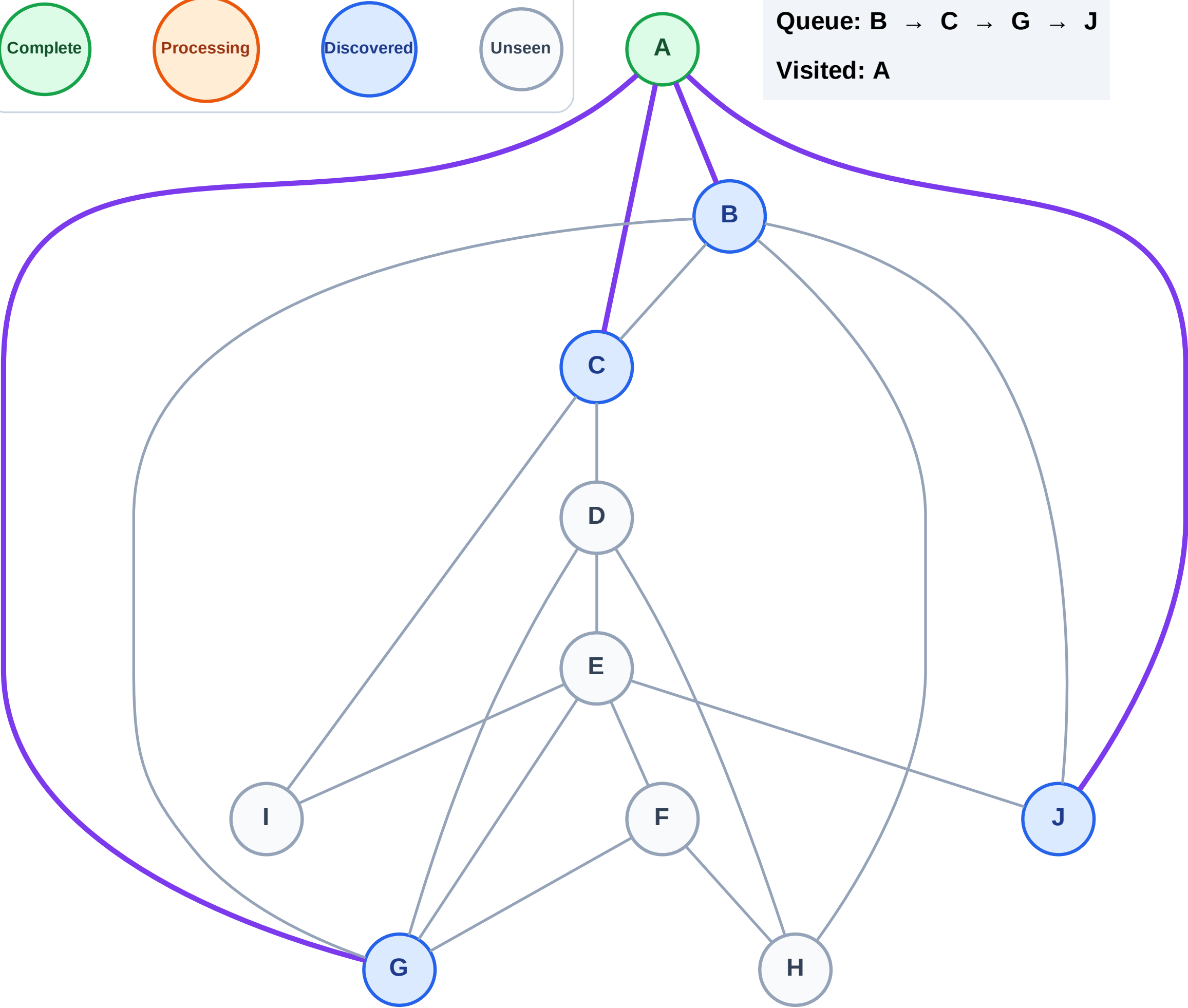
Step 3: Discover B, C, G, J from A

Legend



Queue: B → C → G → J

Visited: A



BFS Traversal

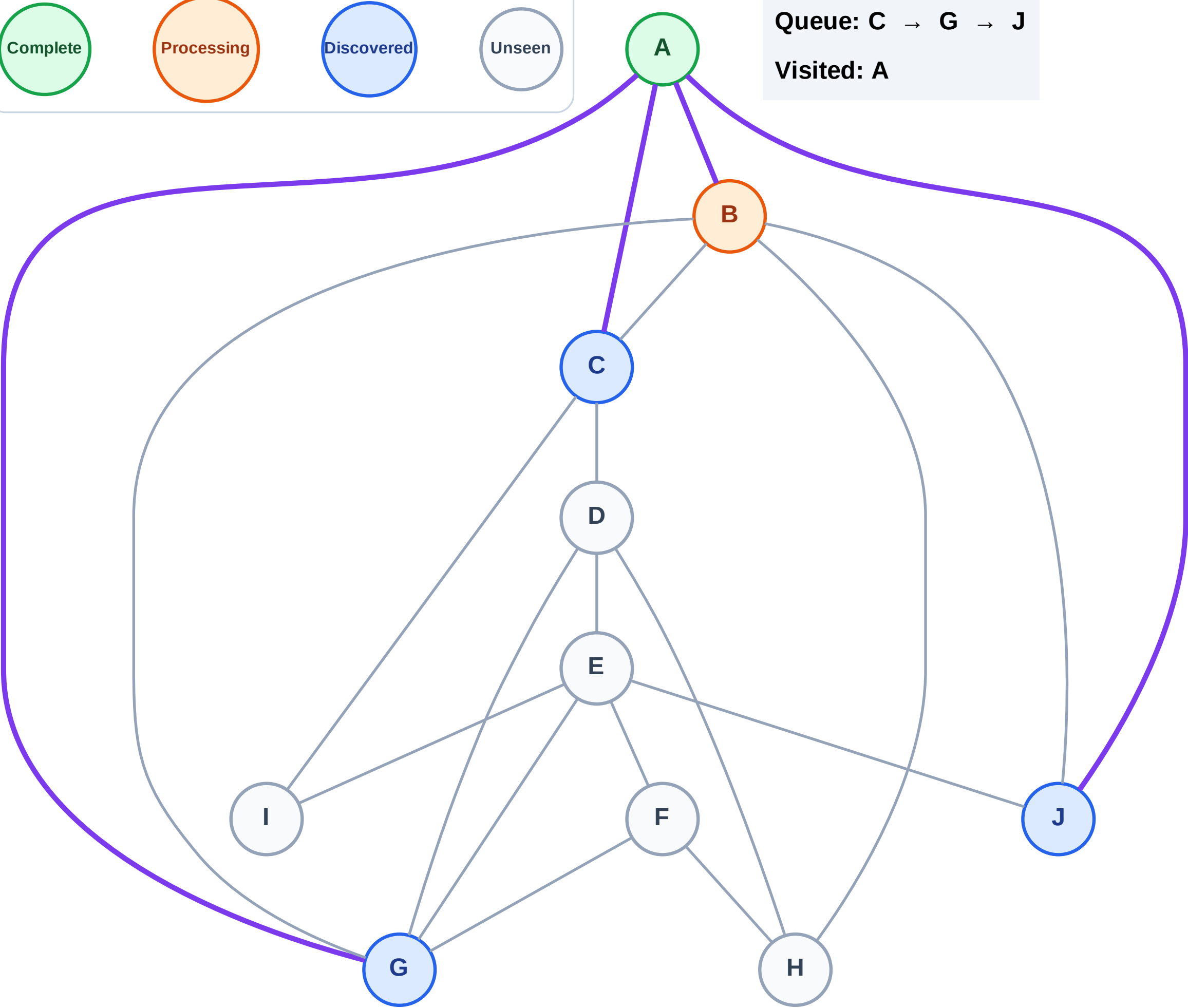
Step 4: Process node B

Legend



Queue: C → G → J

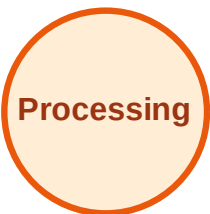
Visited: A



BFS Traversal

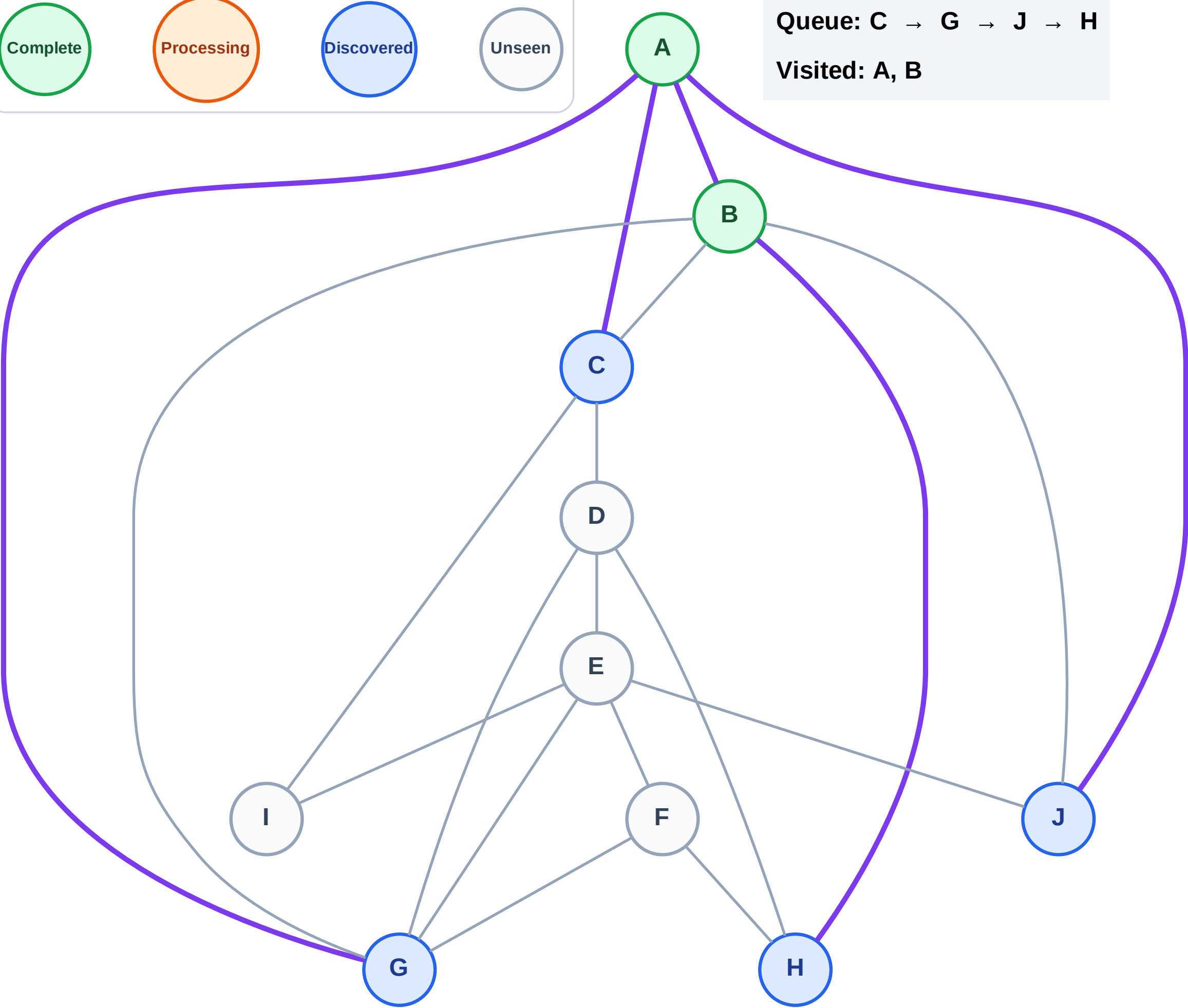
Step 5: Discover H from B

Legend



Queue: C → G → J → H

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend



Complete



Processing



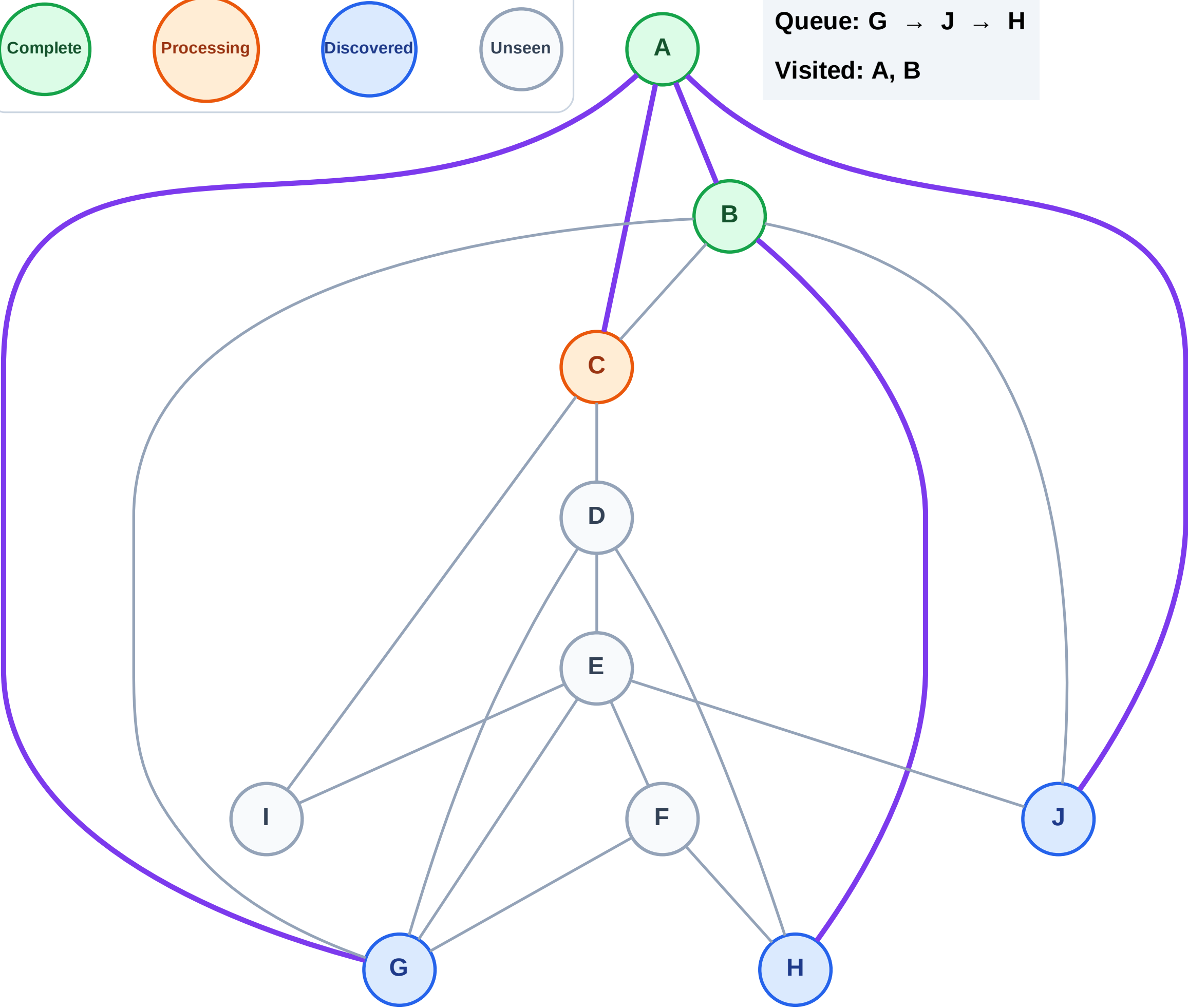
Discovered



Unseen

Queue: G → J → H

Visited: A, B



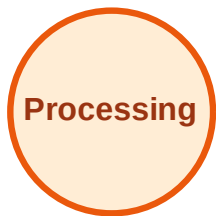
BFS Traversal

Step 7: Discover D, I from C

Legend



Complete



Processing



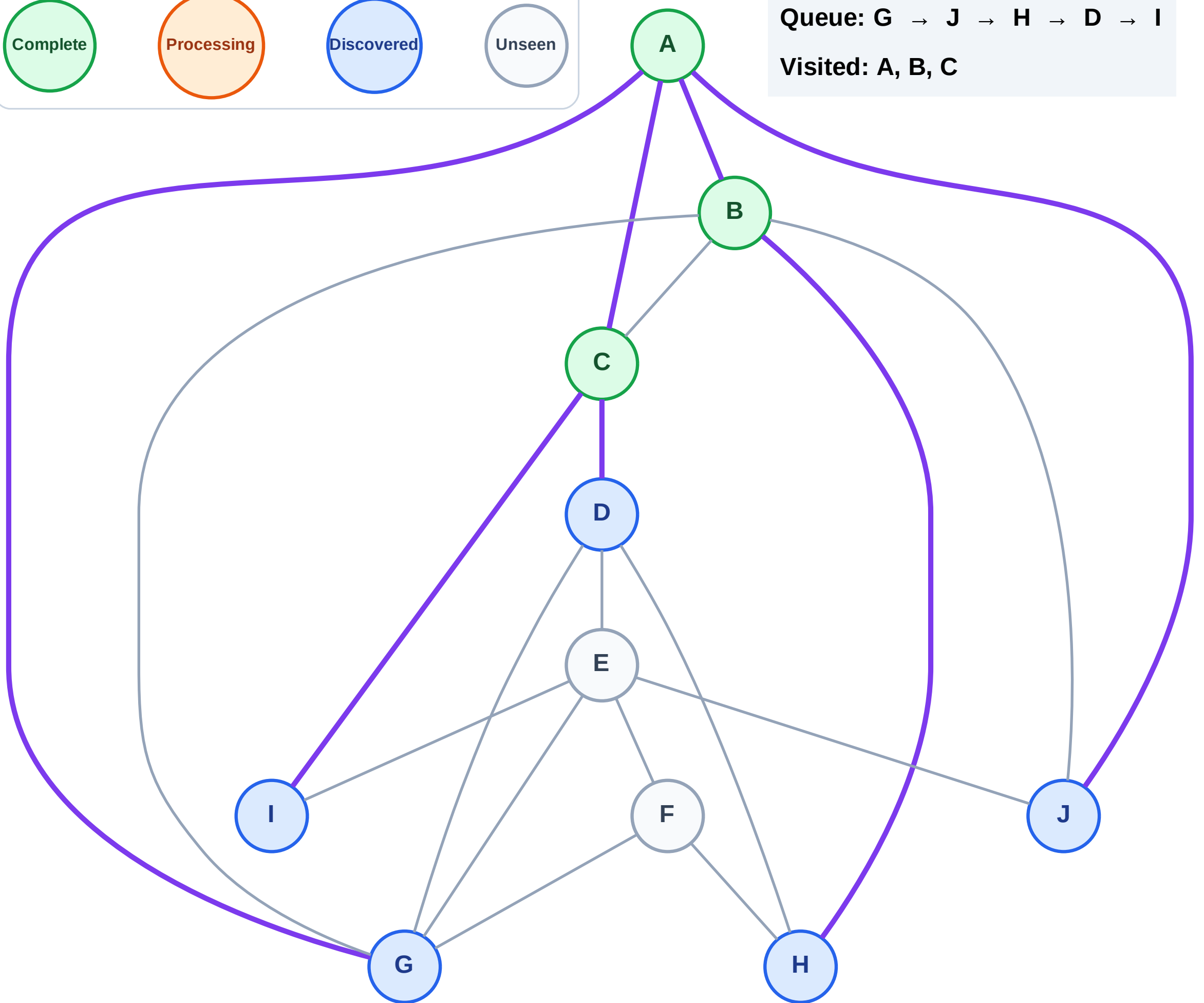
Discovered



Unseen

Queue: G → J → H → D → I

Visited: A, B, C



BFS Traversal

Step 8: Process node G

Legend



Complete



Processing



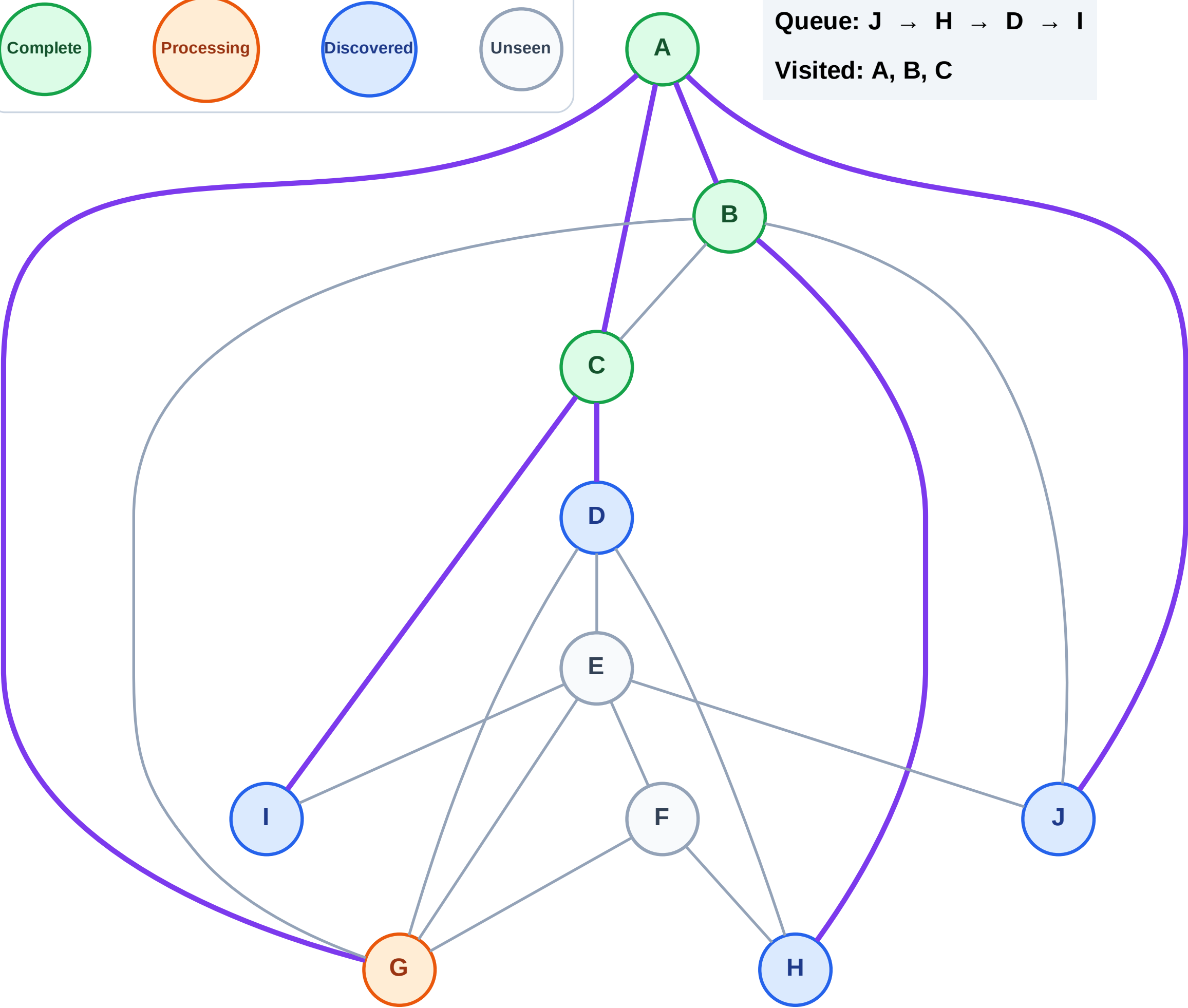
Discovered



Unseen

Queue: J → H → D → I

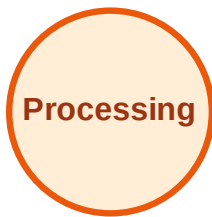
Visited: A, B, C



BFS Traversal

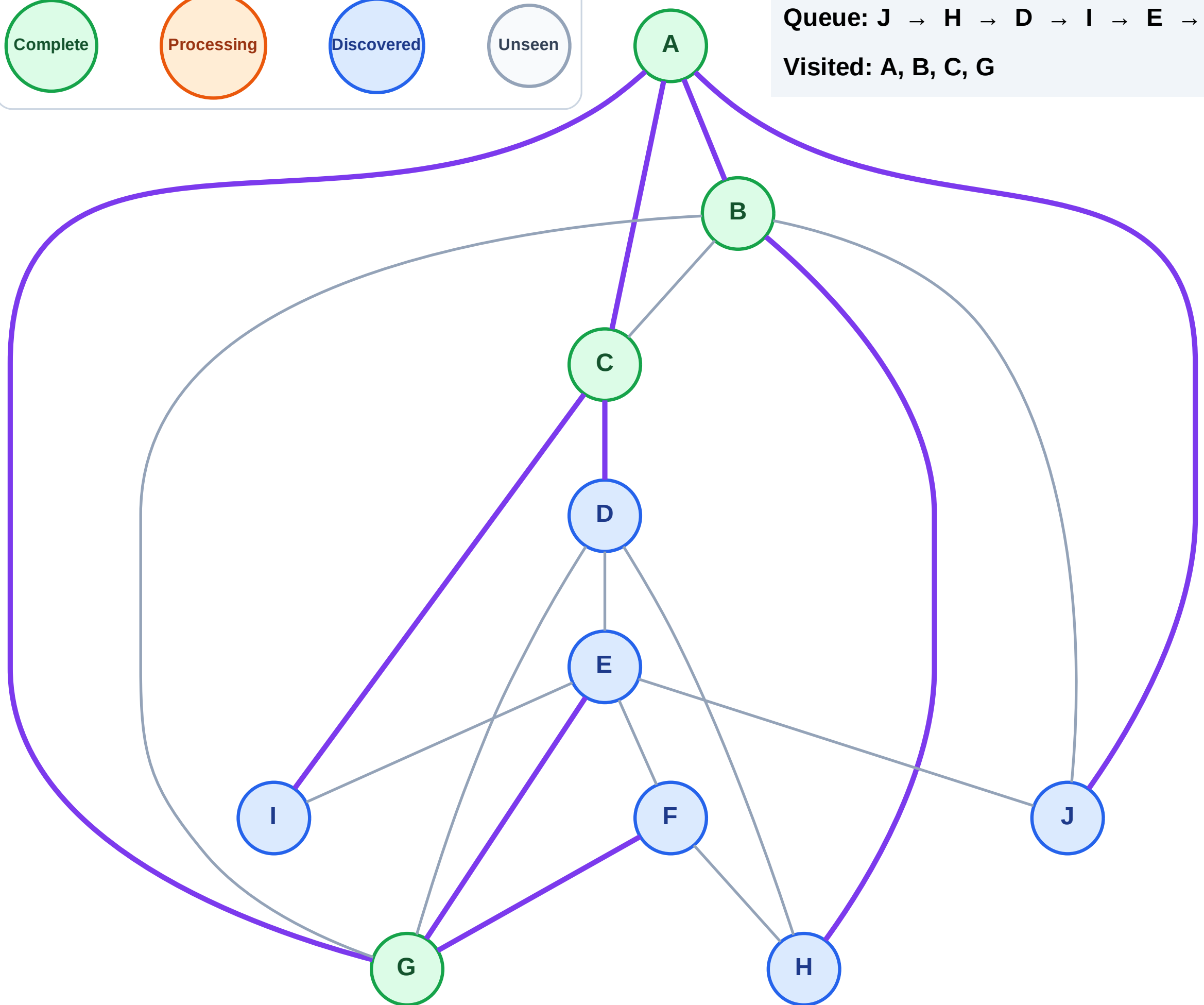
Step 9: Discover E, F from G

Legend



Queue: J → H → D → I → E → F

Visited: A, B, C, G



BFS Traversal

Step 10: Process node J

Legend



Complete



Processing



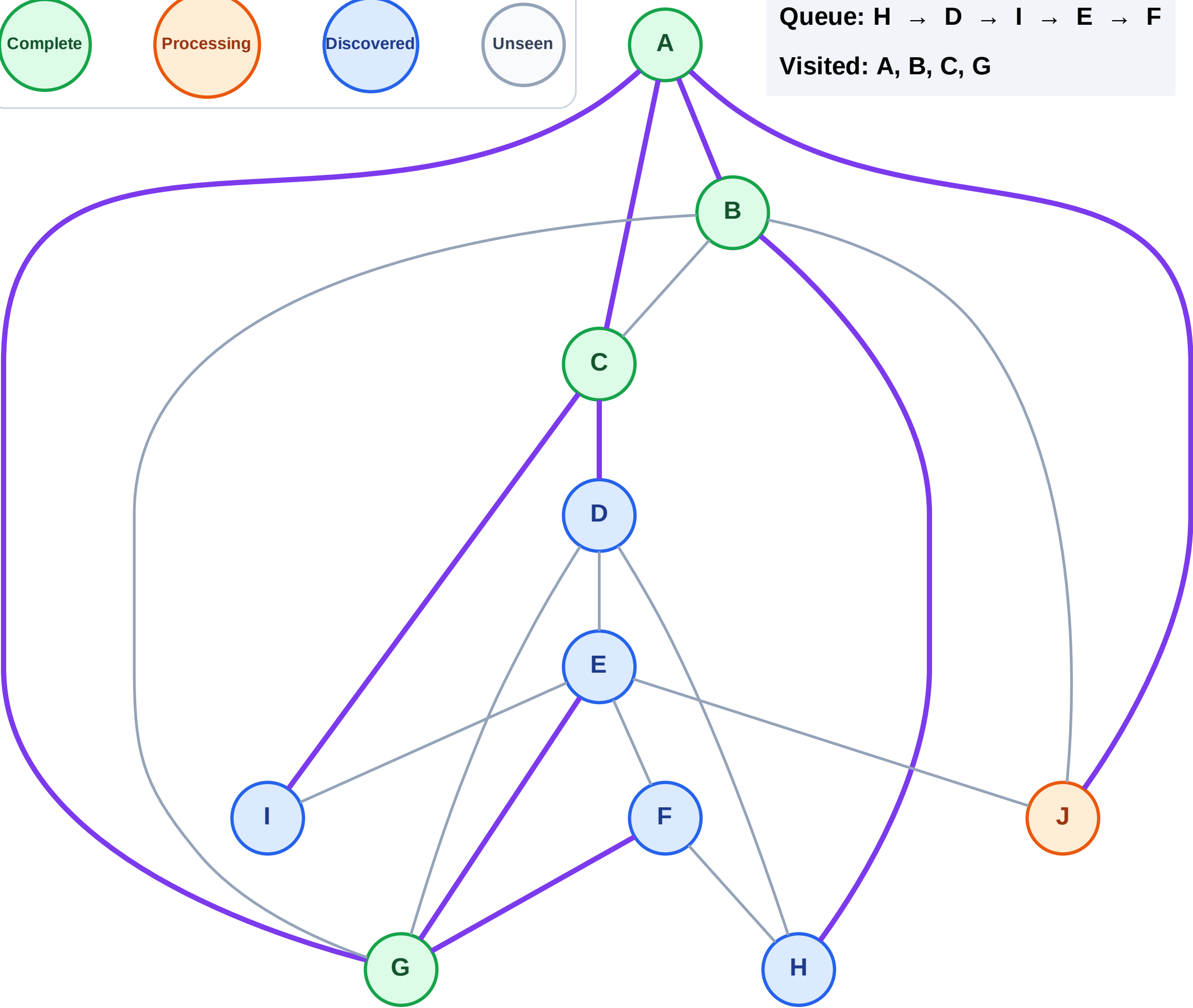
Discovered



Unseen

Queue: H → D → I → E → F

Visited: A, B, C, G



BFS Traversal

Step 11: No new neighbors from J

Legend



Complete



Processing



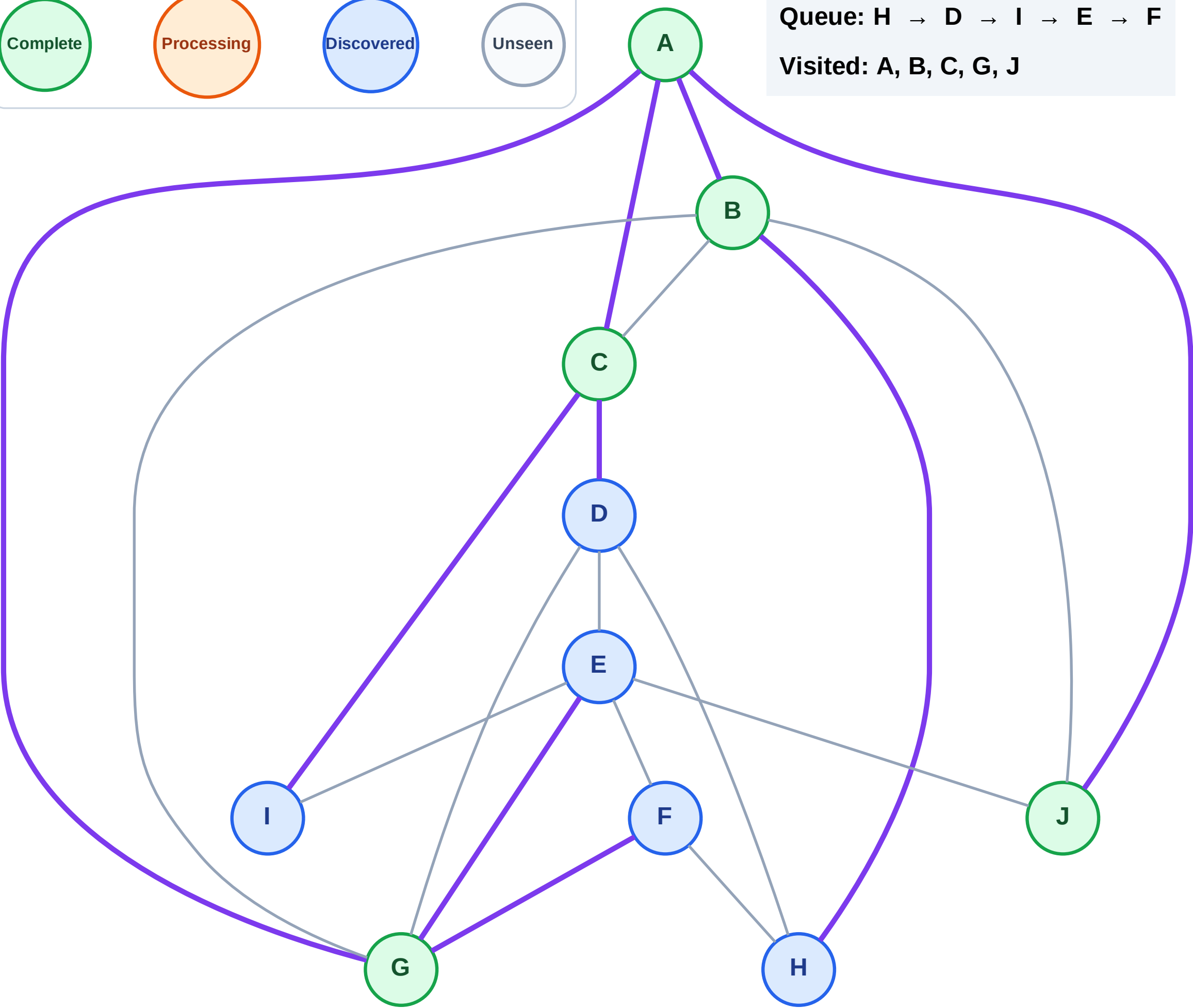
Discovered



Unseen

Queue: H → D → I → E → F

Visited: A, B, C, G, J



BFS Traversal

Step 12: Process node H

Legend



Complete



Processing



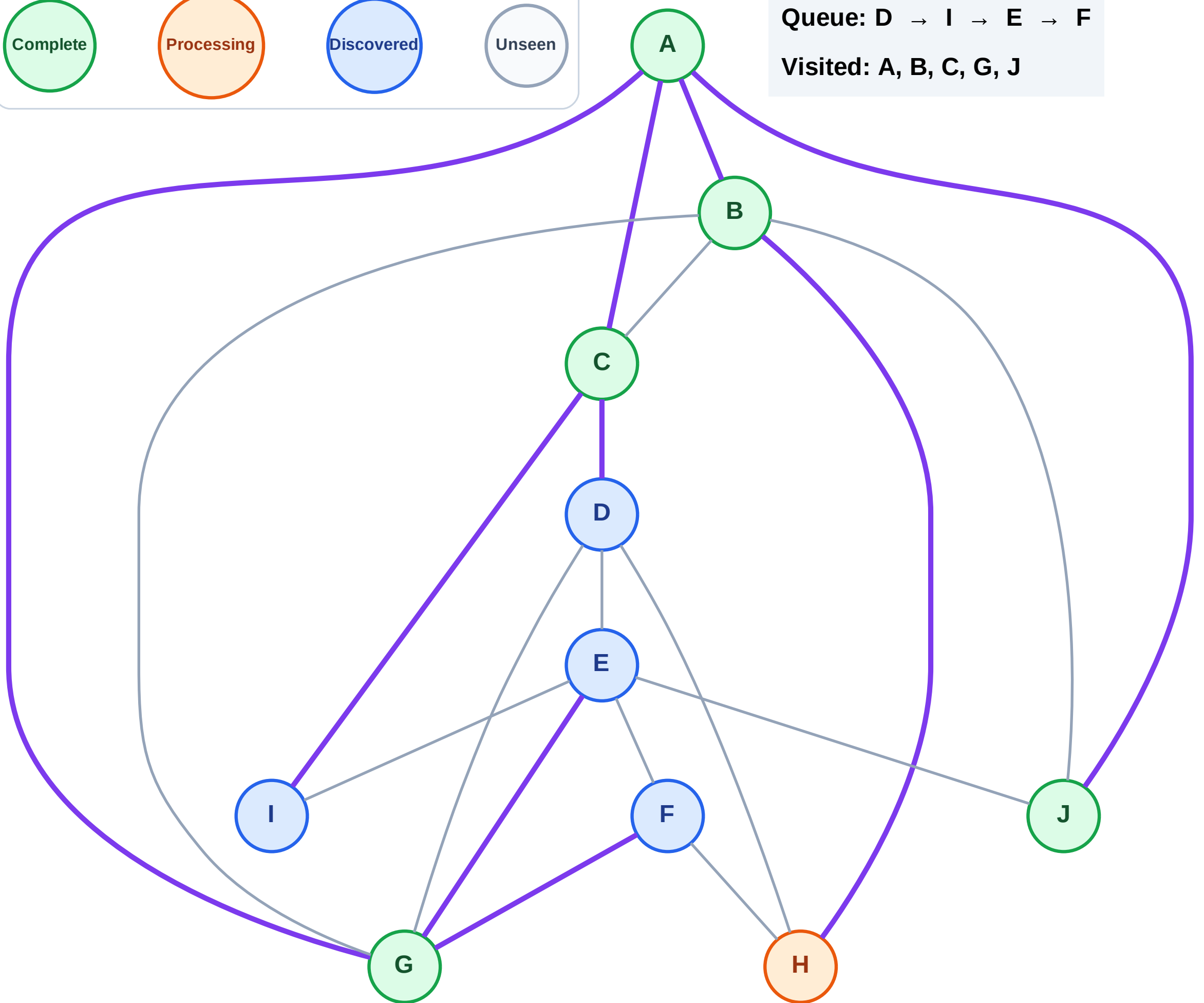
Discovered



Unseen

Queue: D → I → E → F

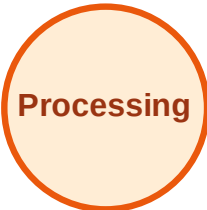
Visited: A, B, C, G, J



BFS Traversal

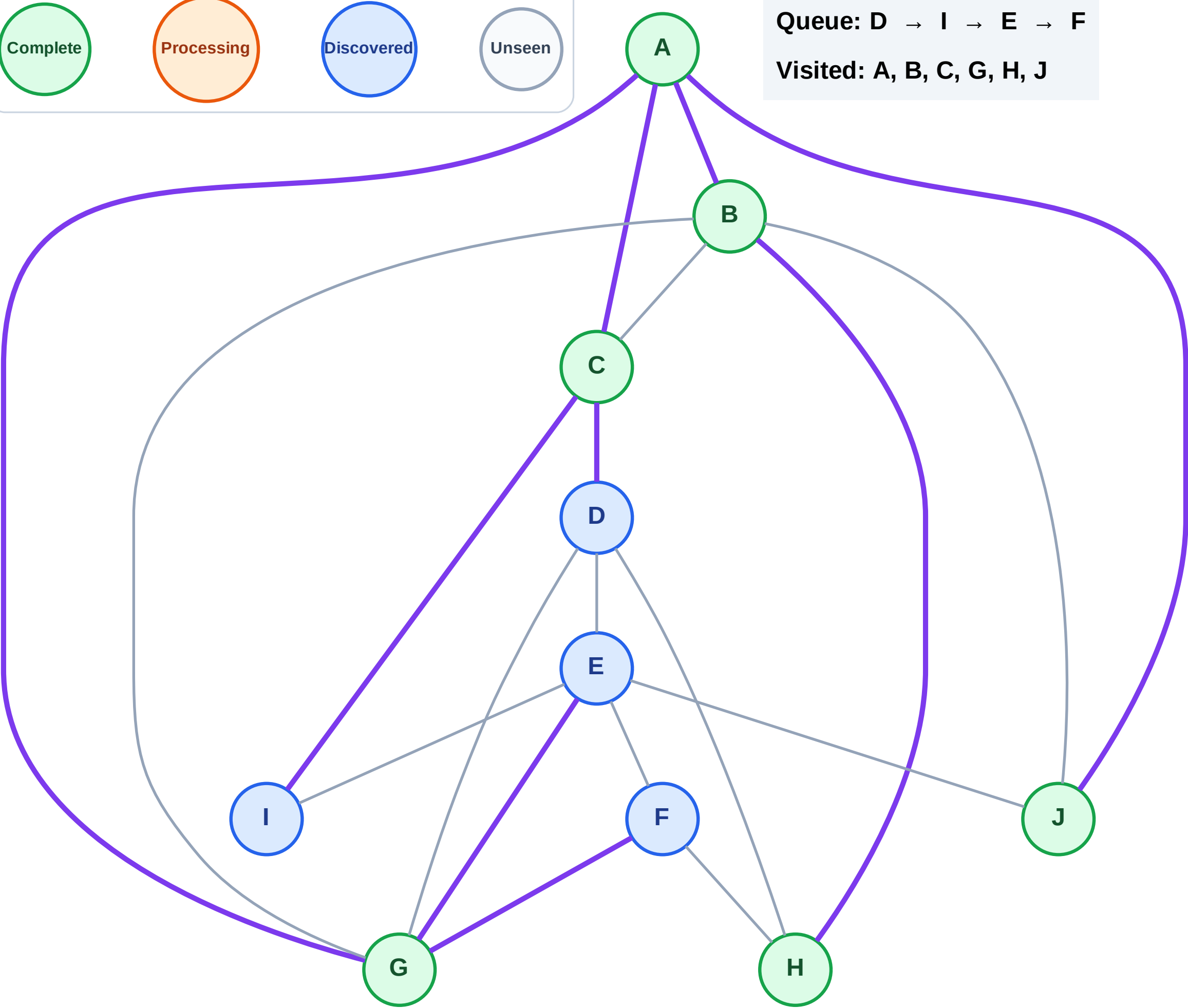
Step 13: No new neighbors from H

Legend



Queue: D → I → E → F

Visited: A, B, C, G, H, J



BFS Traversal

Step 14: Process node D

Legend



Complete



Processing



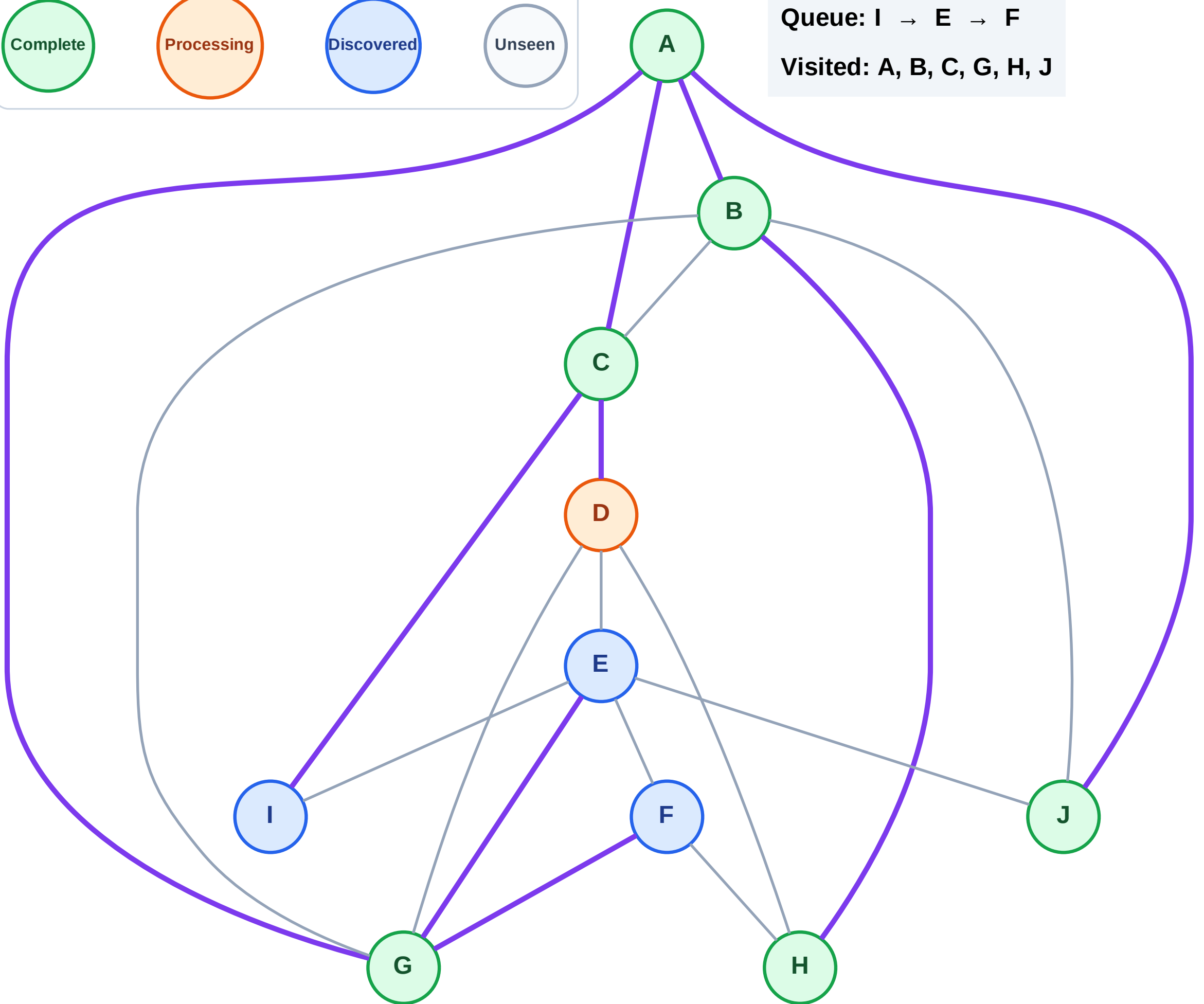
Discovered



Unseen

Queue: I → E → F

Visited: A, B, C, G, H, J



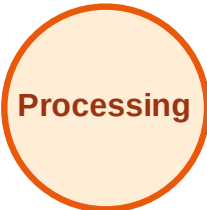
BFS Traversal

Step 15: No new neighbors from D

Legend



Complete



Processing



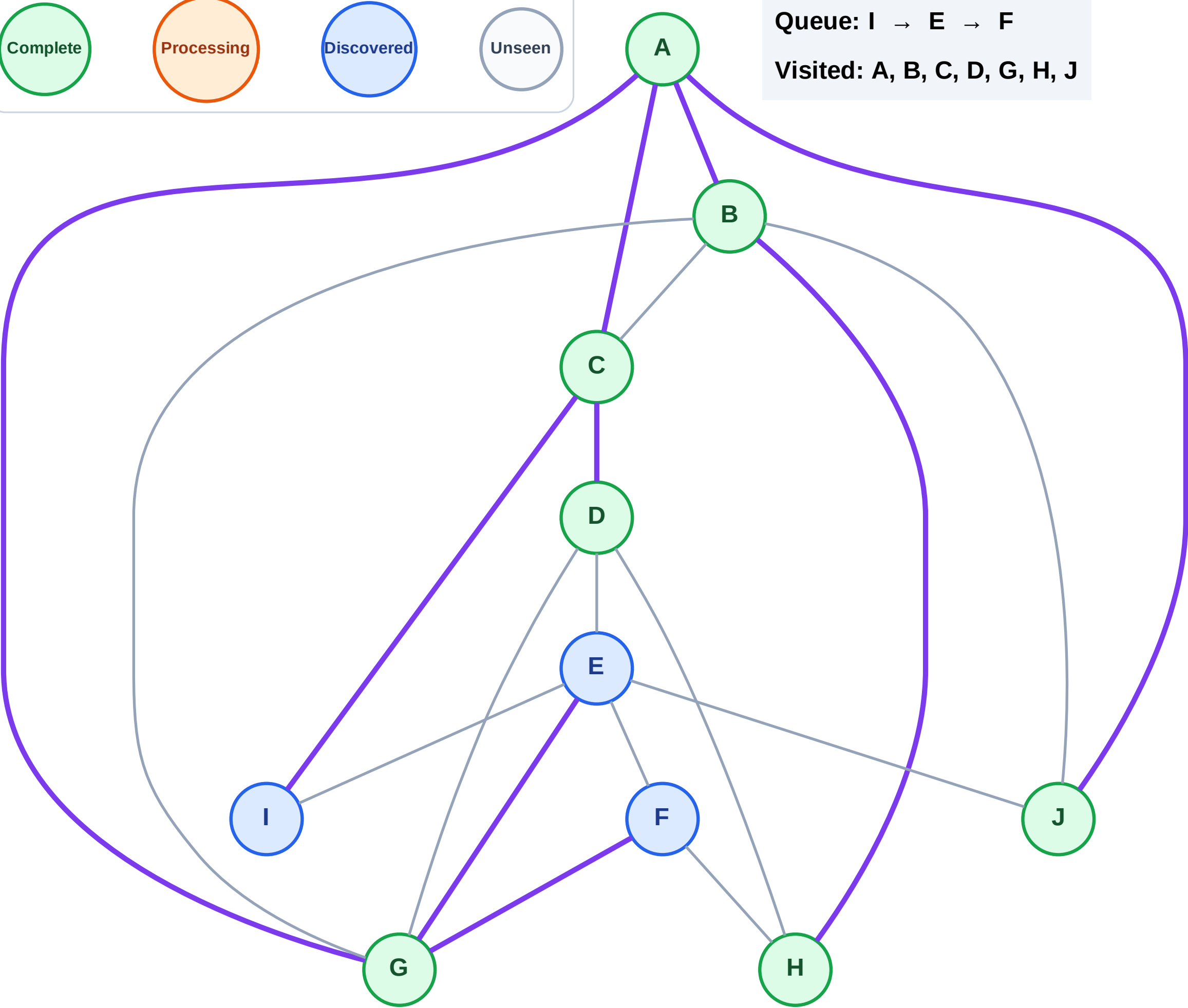
Discovered



Unseen

Queue: I → E → F

Visited: A, B, C, D, G, H, J



BFS Traversal

Step 16: Process node I

Legend



Complete



Processing



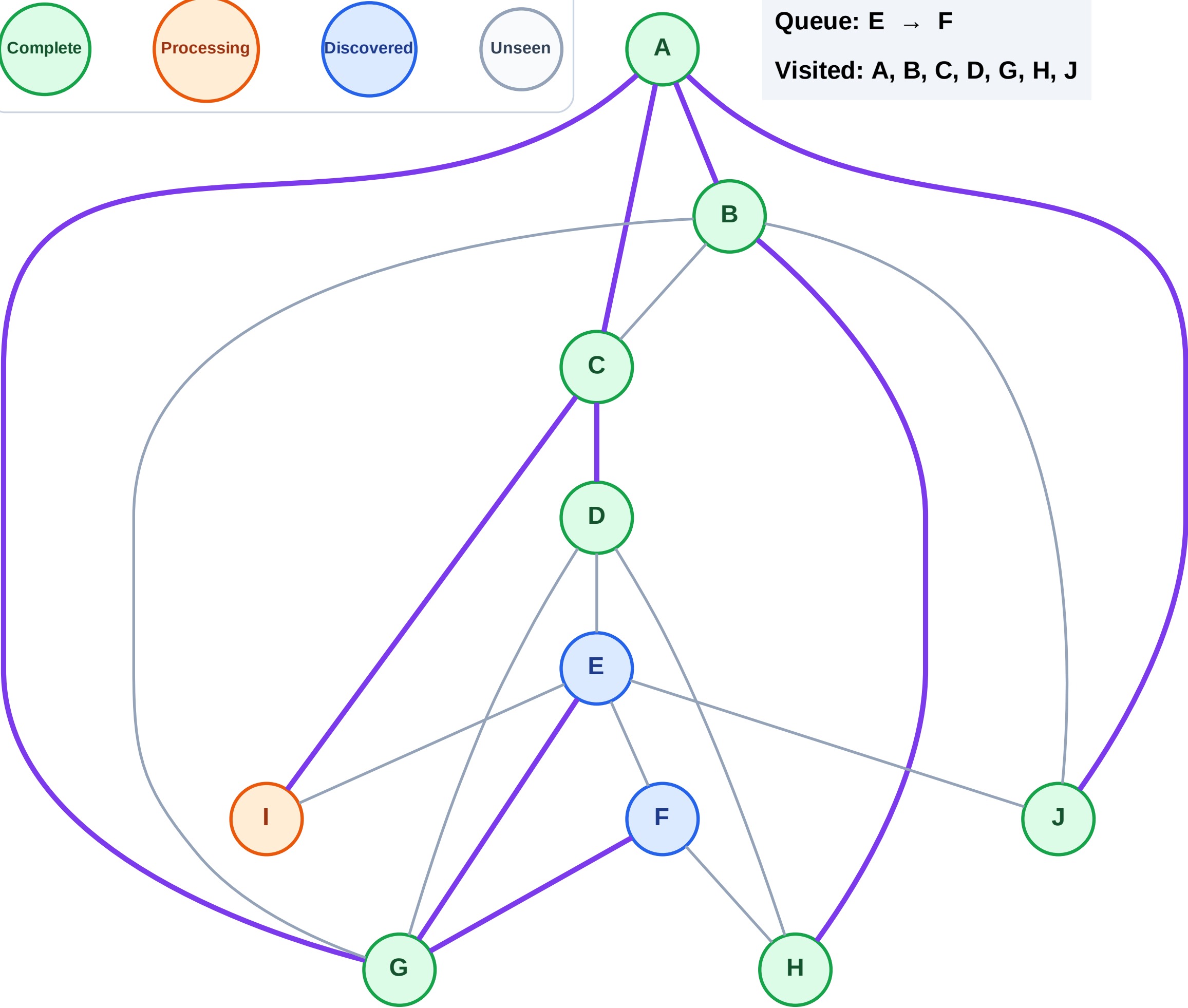
Discovered



Unseen

Queue: E → F

Visited: A, B, C, D, G, H, J



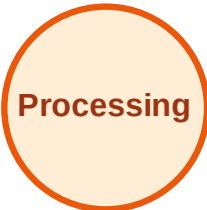
BFS Traversal

Step 17: No new neighbors from I

Legend



Complete



Processing



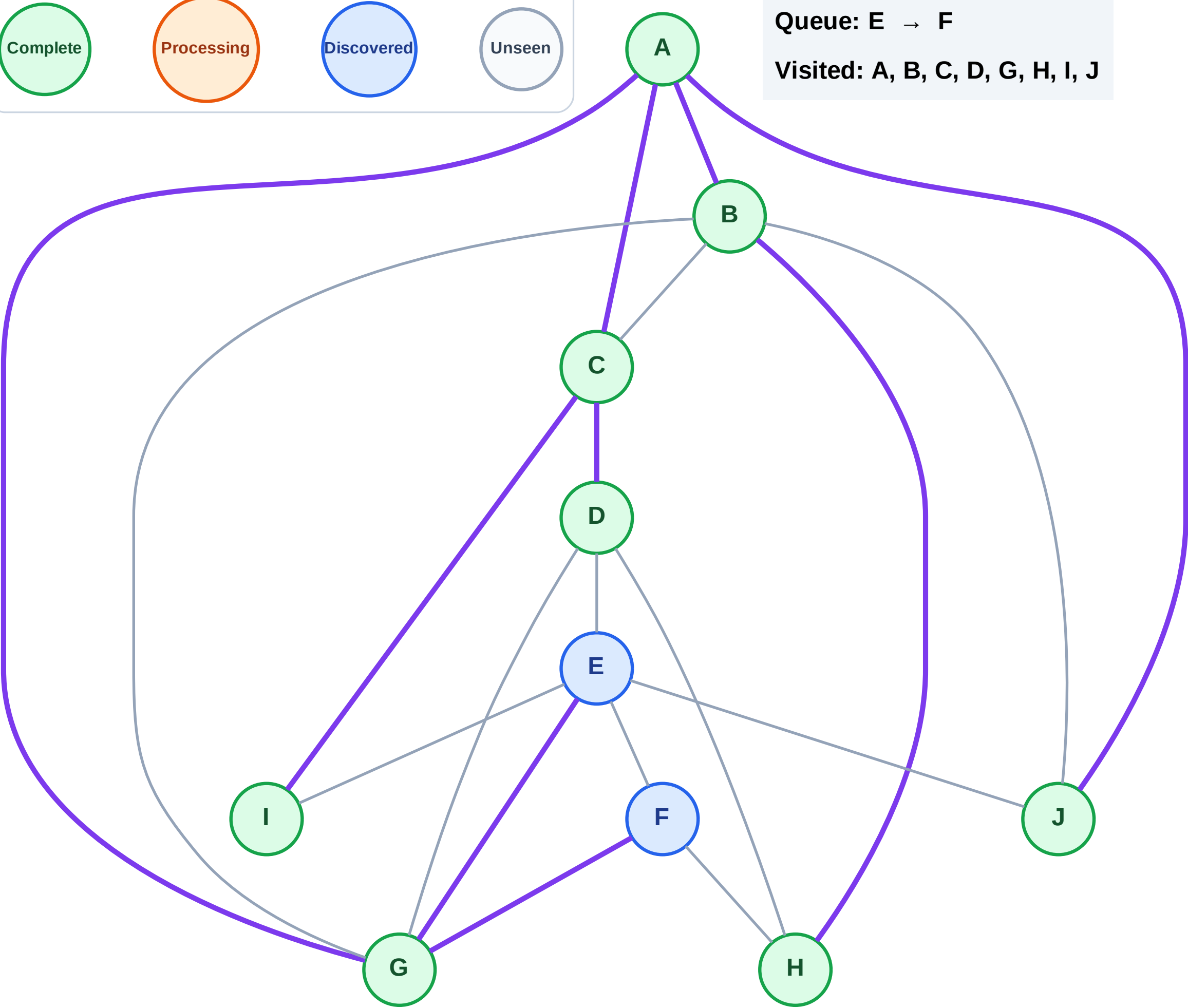
Discovered



Unseen

Queue: E → F

Visited: A, B, C, D, G, H, I, J



BFS Traversal

Step 18: Process node E

Legend



Complete



Processing



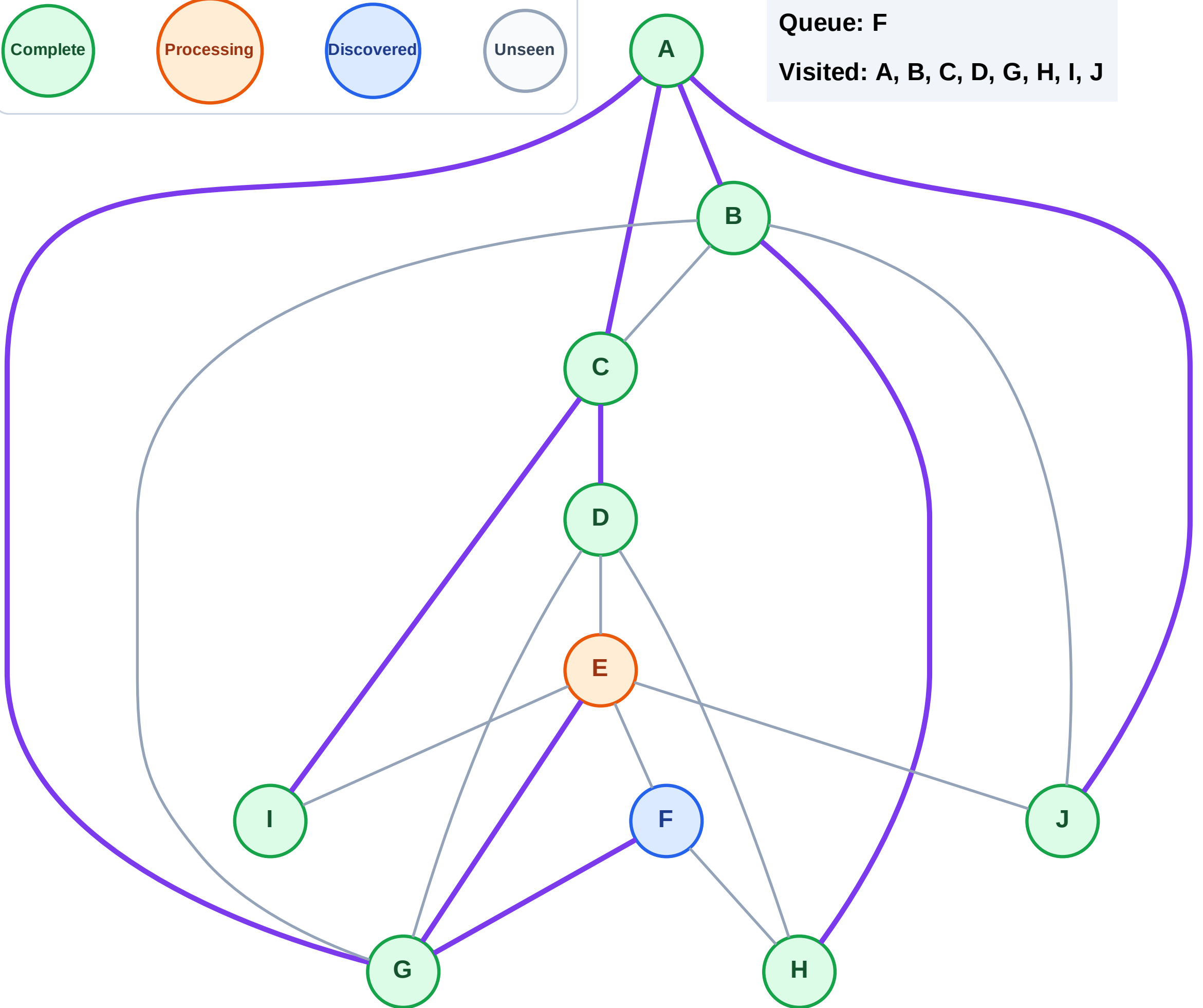
Discovered



Unseen

Queue: F

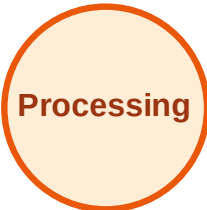
Visited: A, B, C, D, G, H, I, J



BFS Traversal

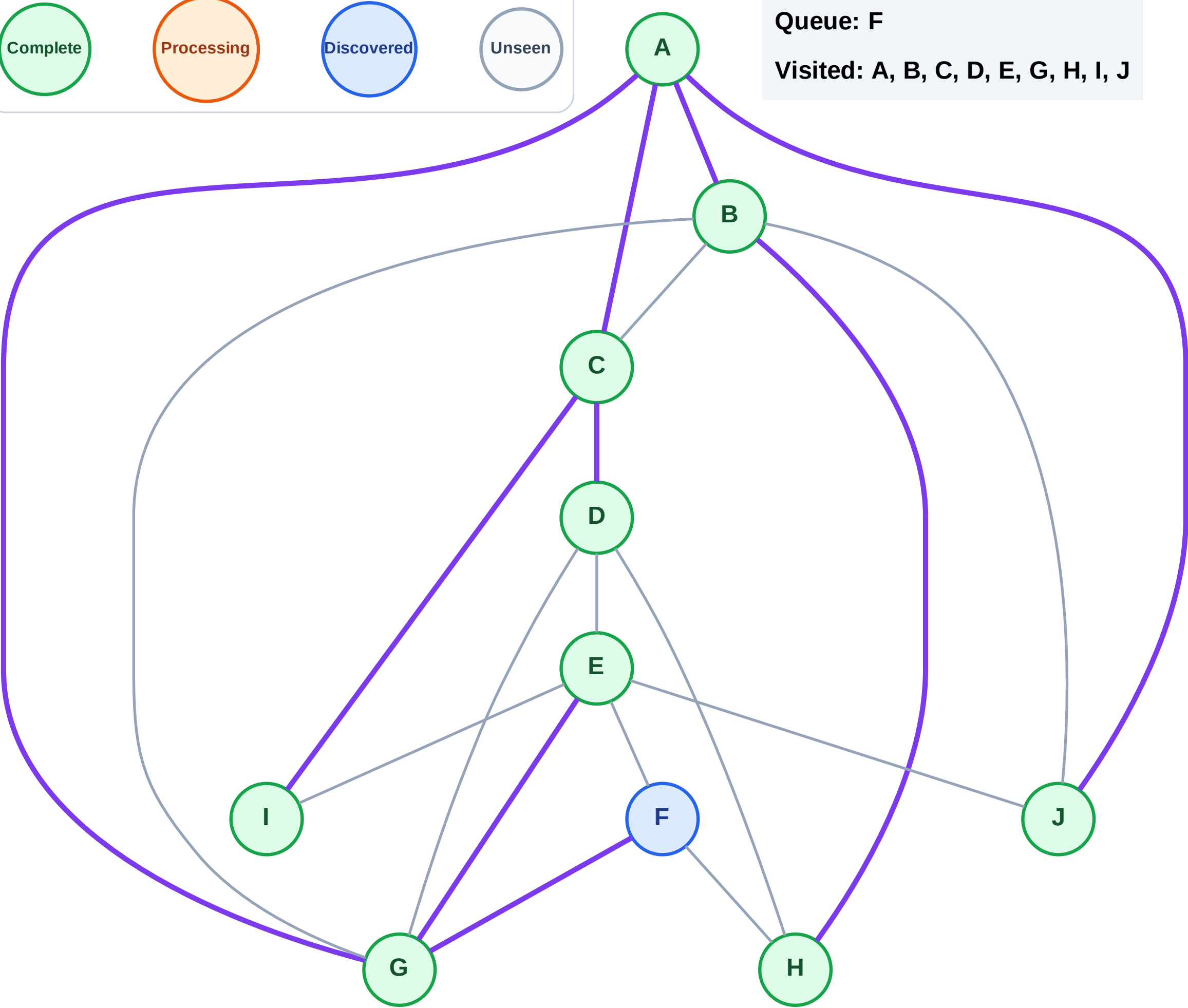
Step 19: No new neighbors from E

Legend



Queue: F

Visited: A, B, C, D, E, G, H, I, J



BFS Traversal

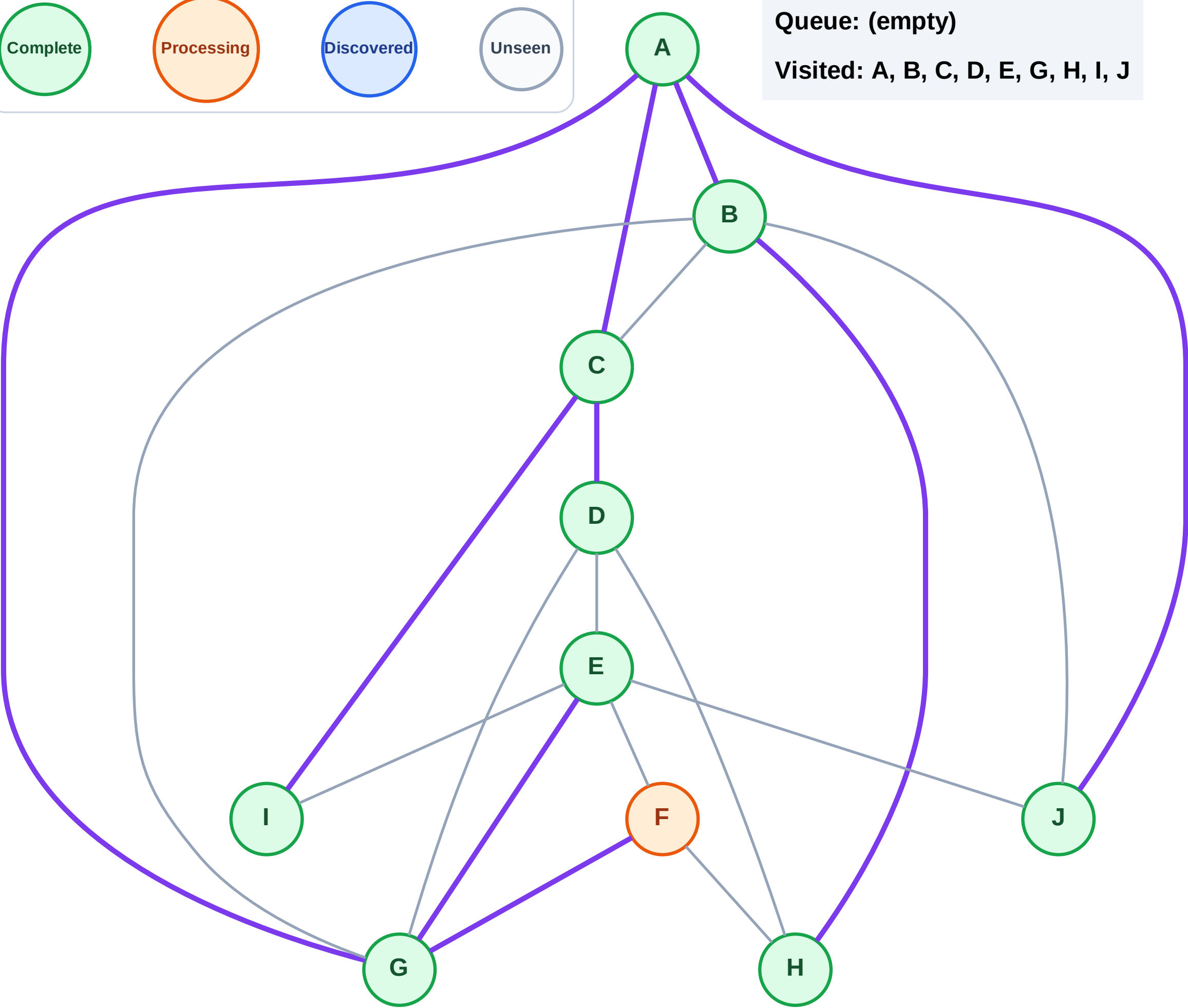
Step 20: Process node F

Legend



Queue: (empty)

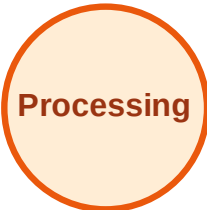
Visited: A, B, C, D, E, G, H, I, J



BFS Traversal

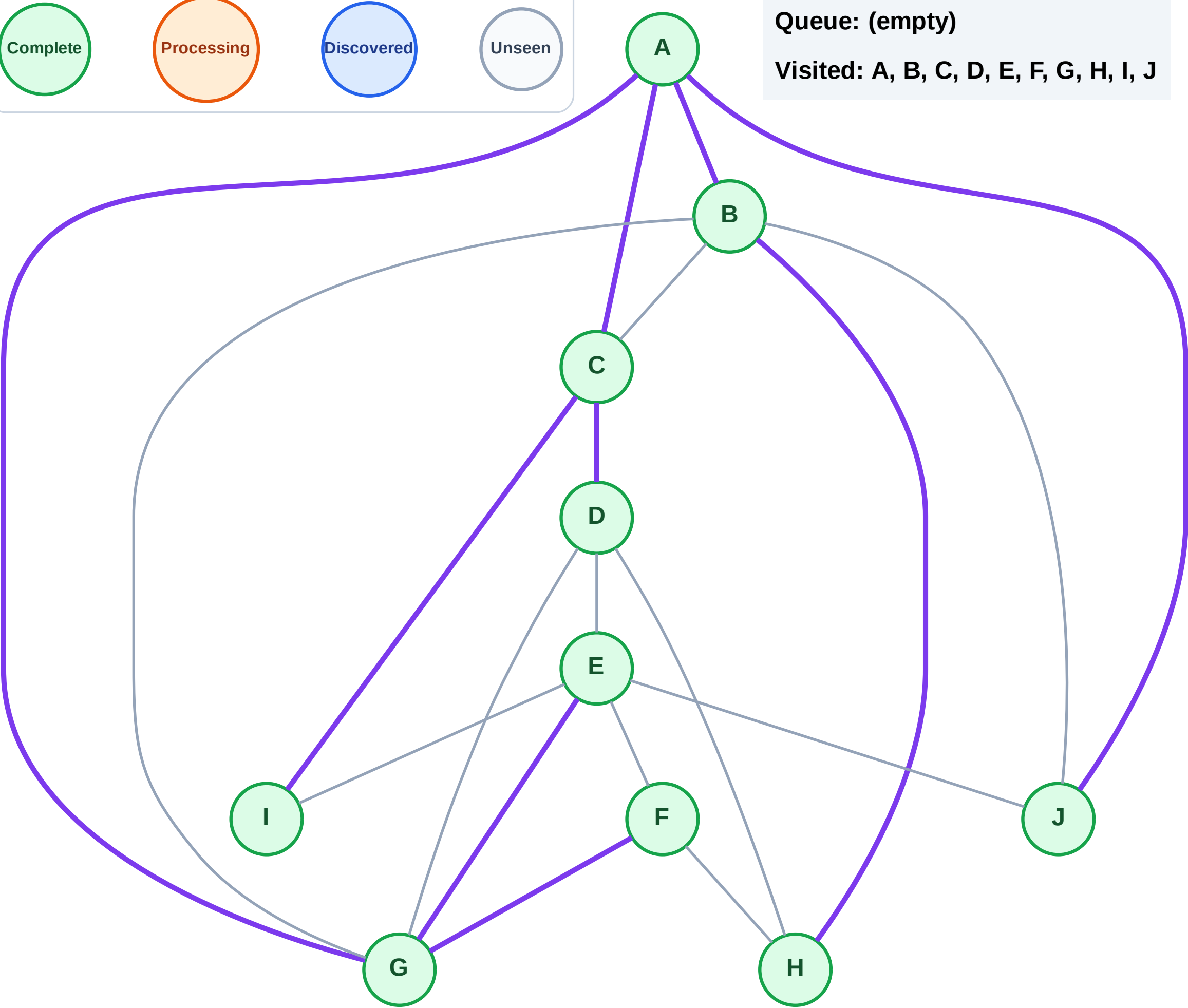
Step 21: No new neighbors from F

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

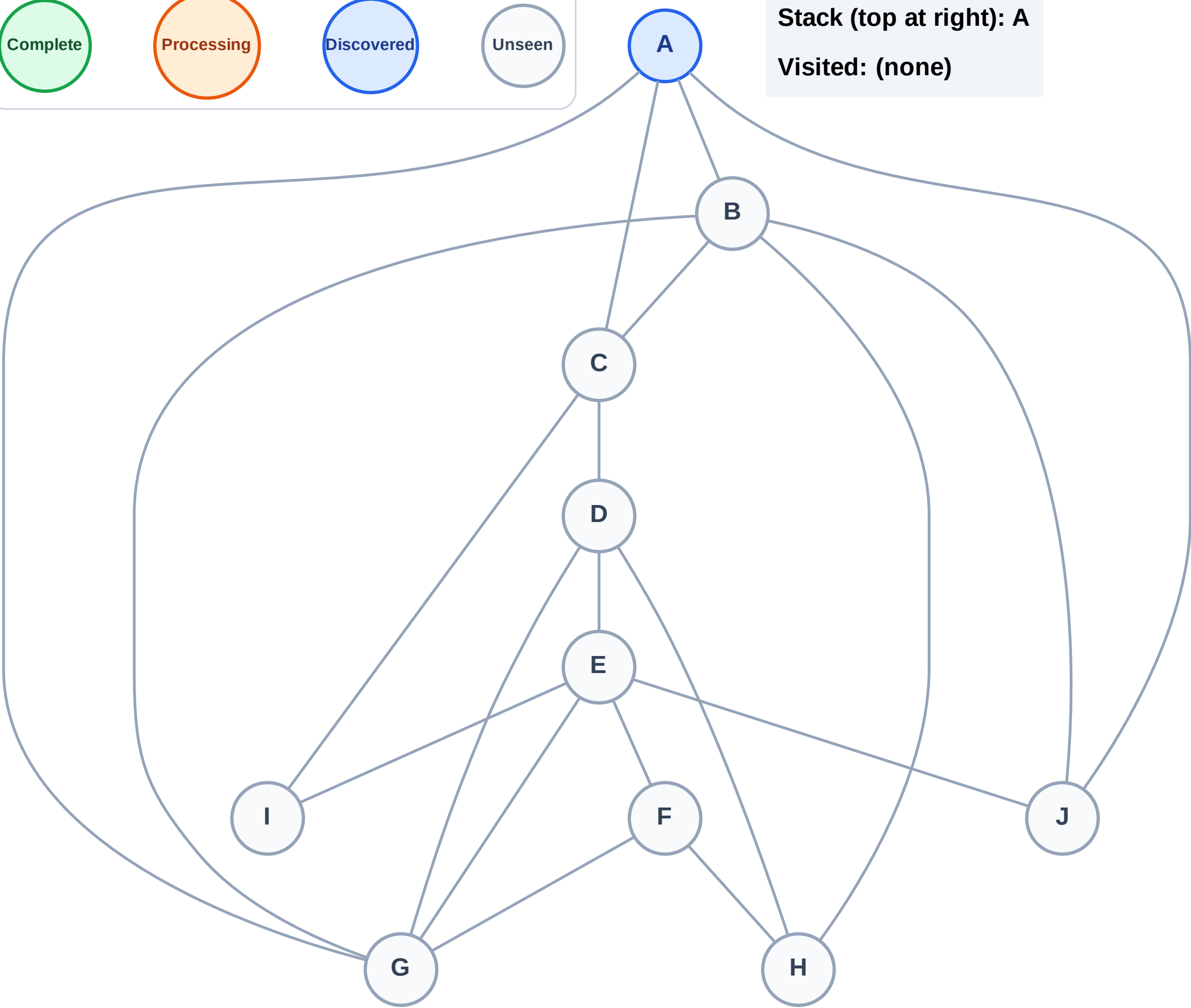
Step 1: Start at A

Legend



Stack (top at right): A

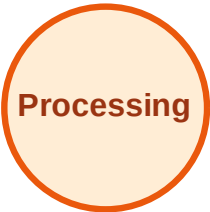
Visited: (none)



DFS Traversal

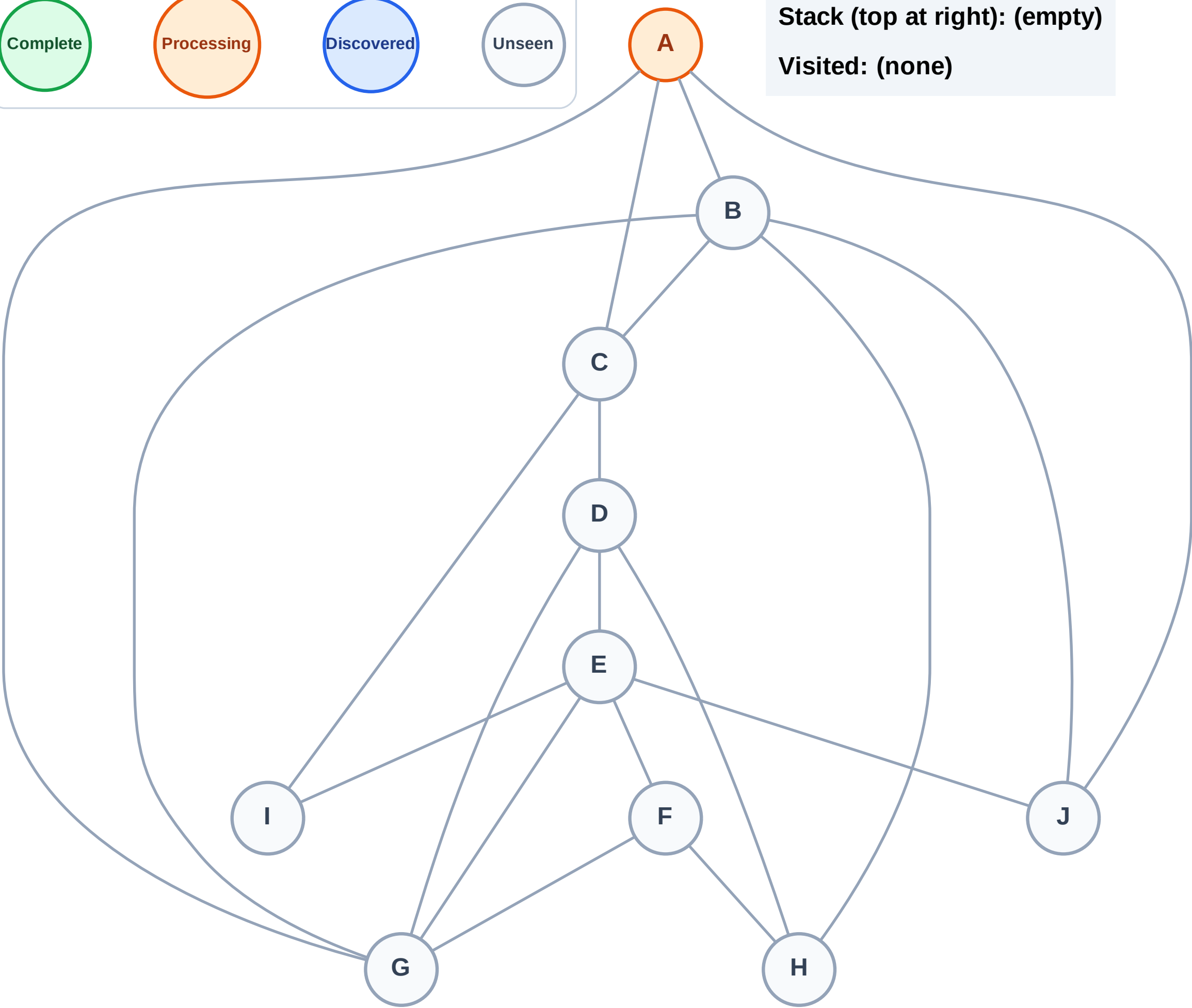
Step 2: Process node A

Legend



Stack (top at right): (empty)

Visited: (none)



DFS Traversal

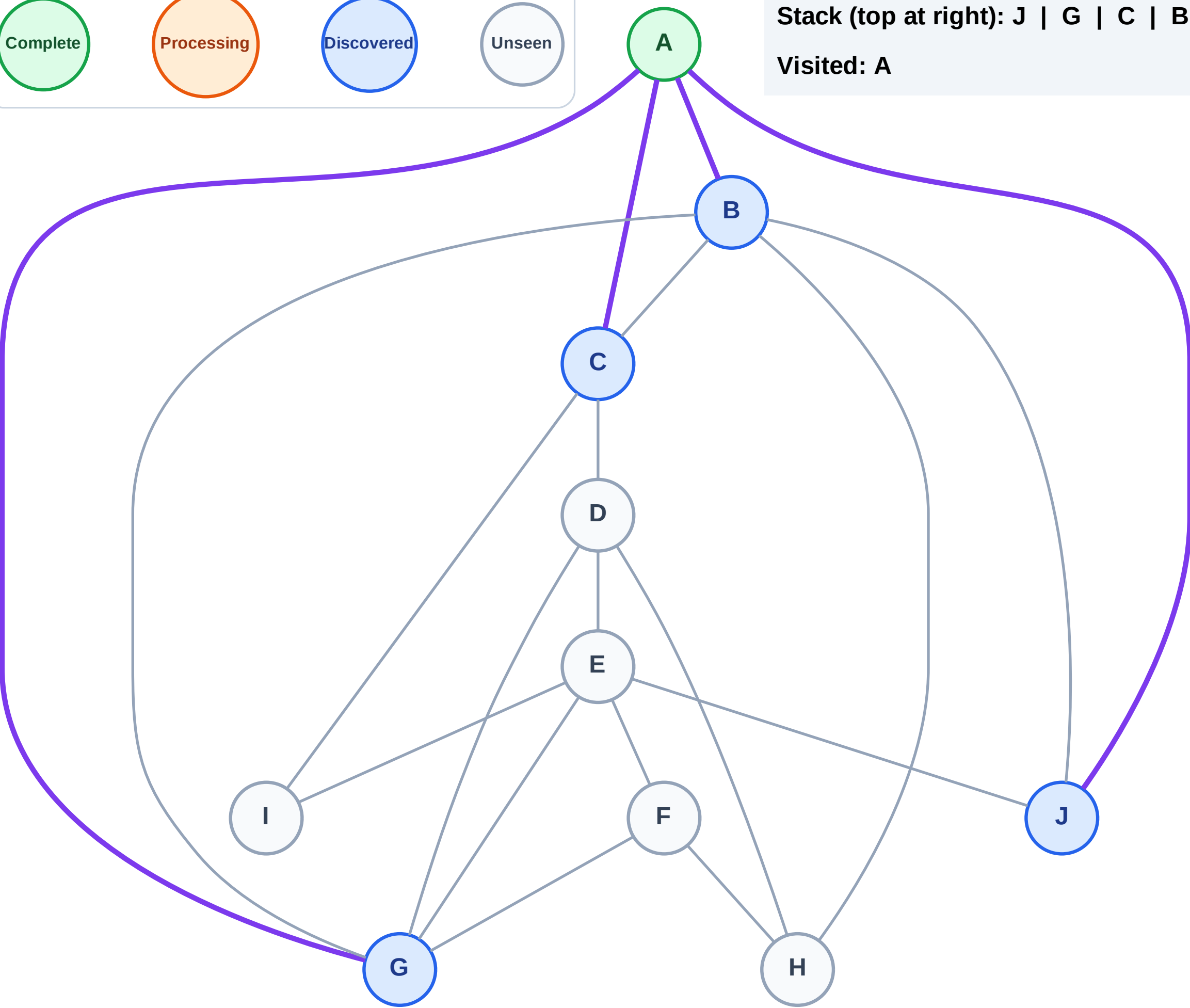
Step 3: Discover J, G, C, B from A

Legend



Stack (top at right): J | G | C | B

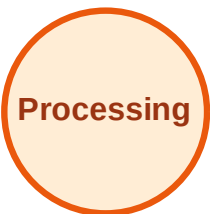
Visited: A



DFS Traversal

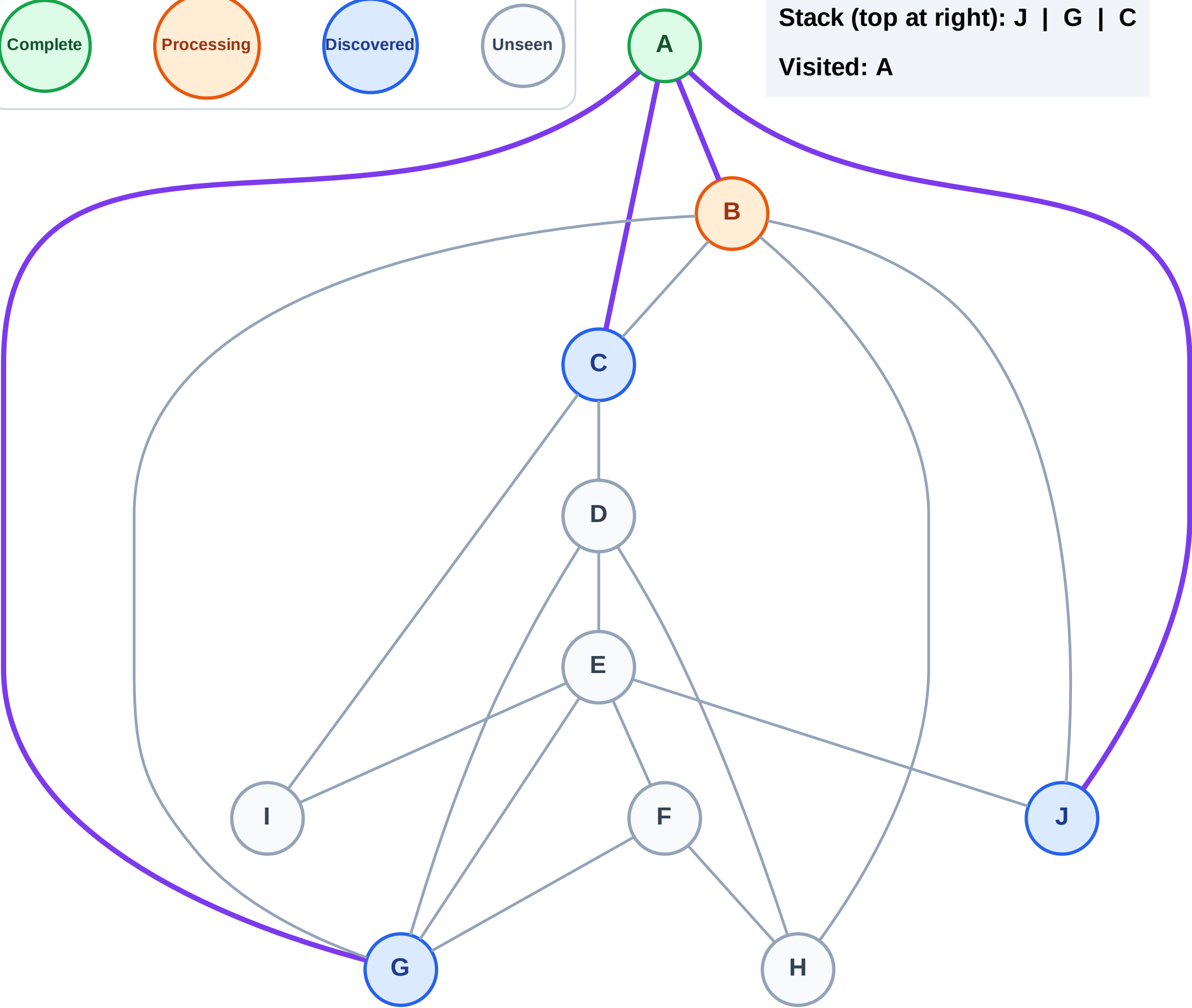
Step 4: Process node B

Legend

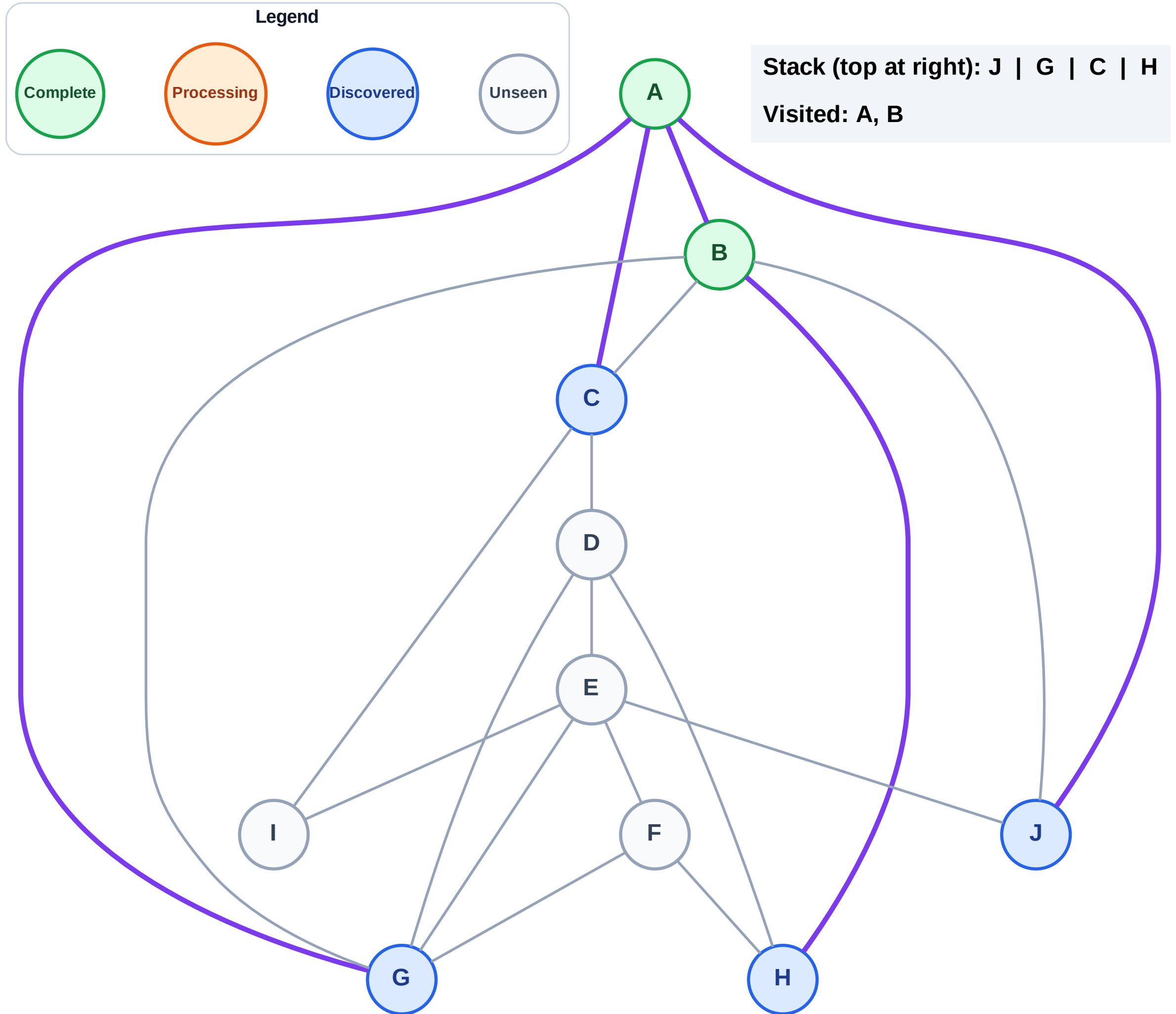


Stack (top at right): J | G | C

Visited: A



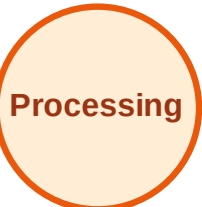
Step 5: Discover H from B



DFS Traversal

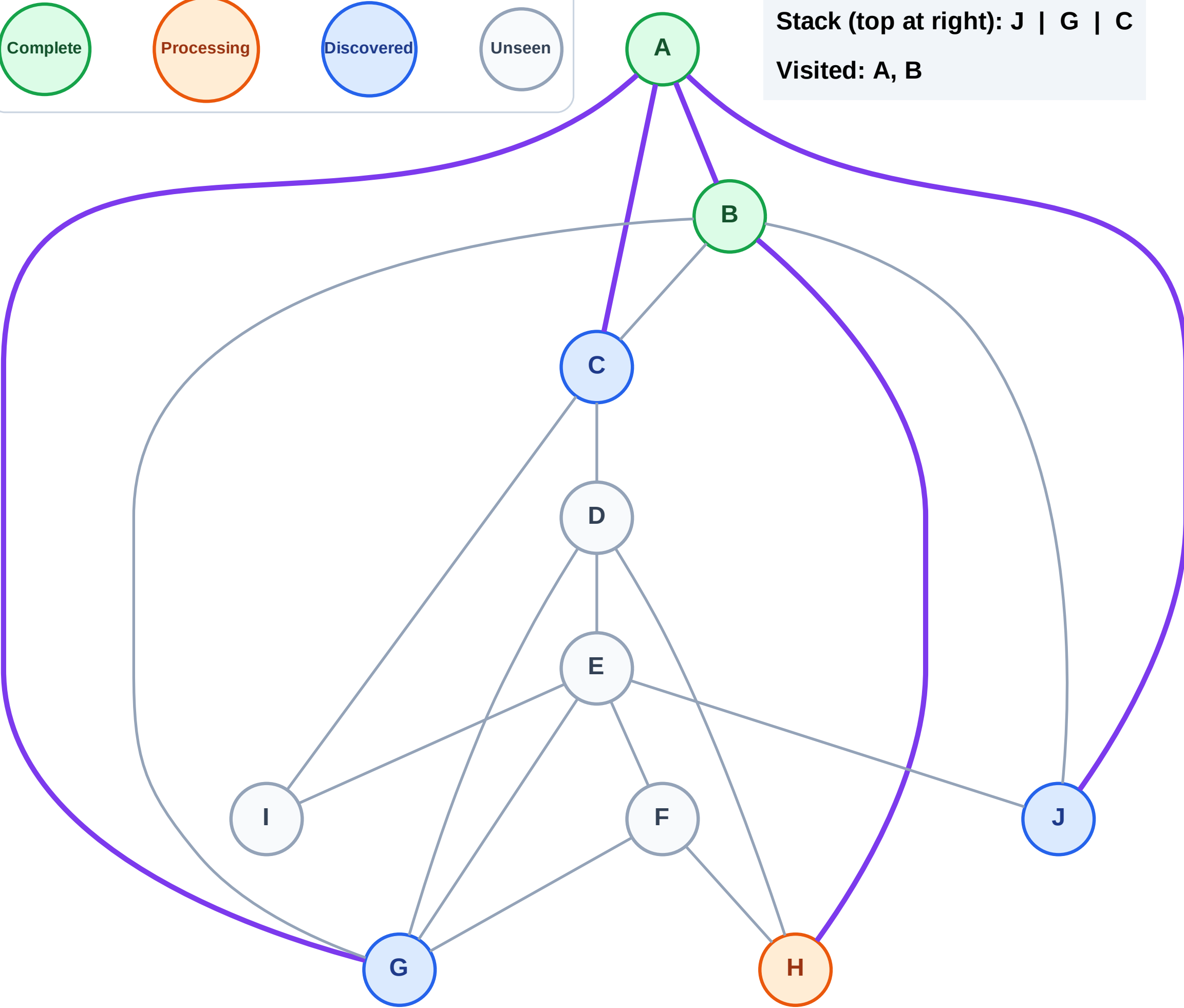
Step 6: Process node H

Legend



Stack (top at right): J | G | C

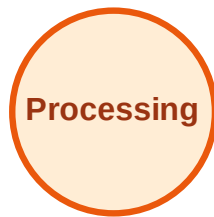
Visited: A, B



DFS Traversal

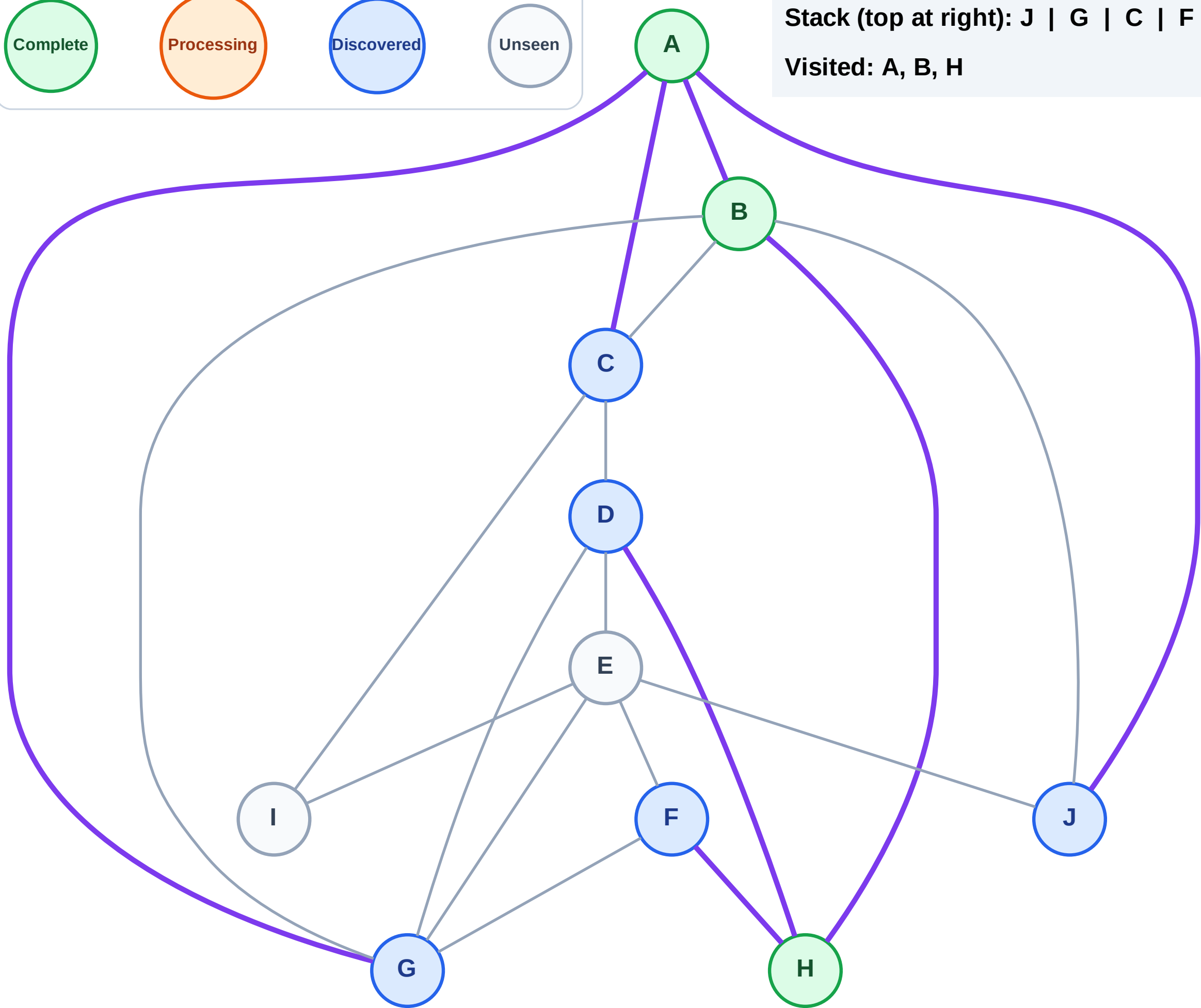
Step 7: Discover F, D from H

Legend



Stack (top at right): J | G | C | F | D

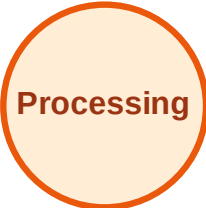
Visited: A, B, H



DFS Traversal

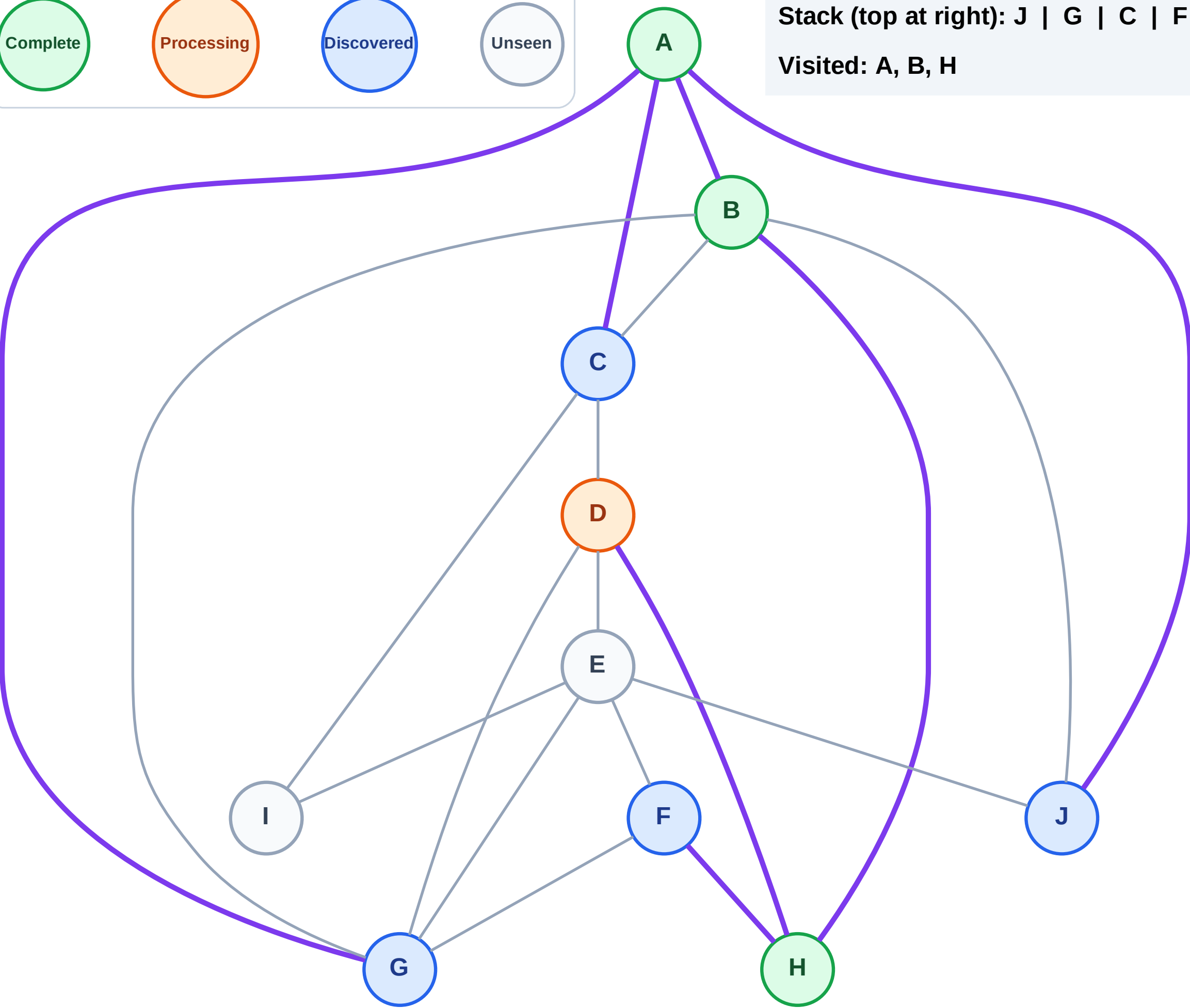
Step 8: Process node D

Legend



Stack (top at right): J | G | C | F

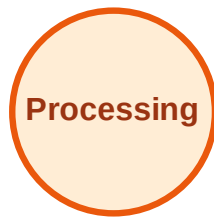
Visited: A, B, H



DFS Traversal

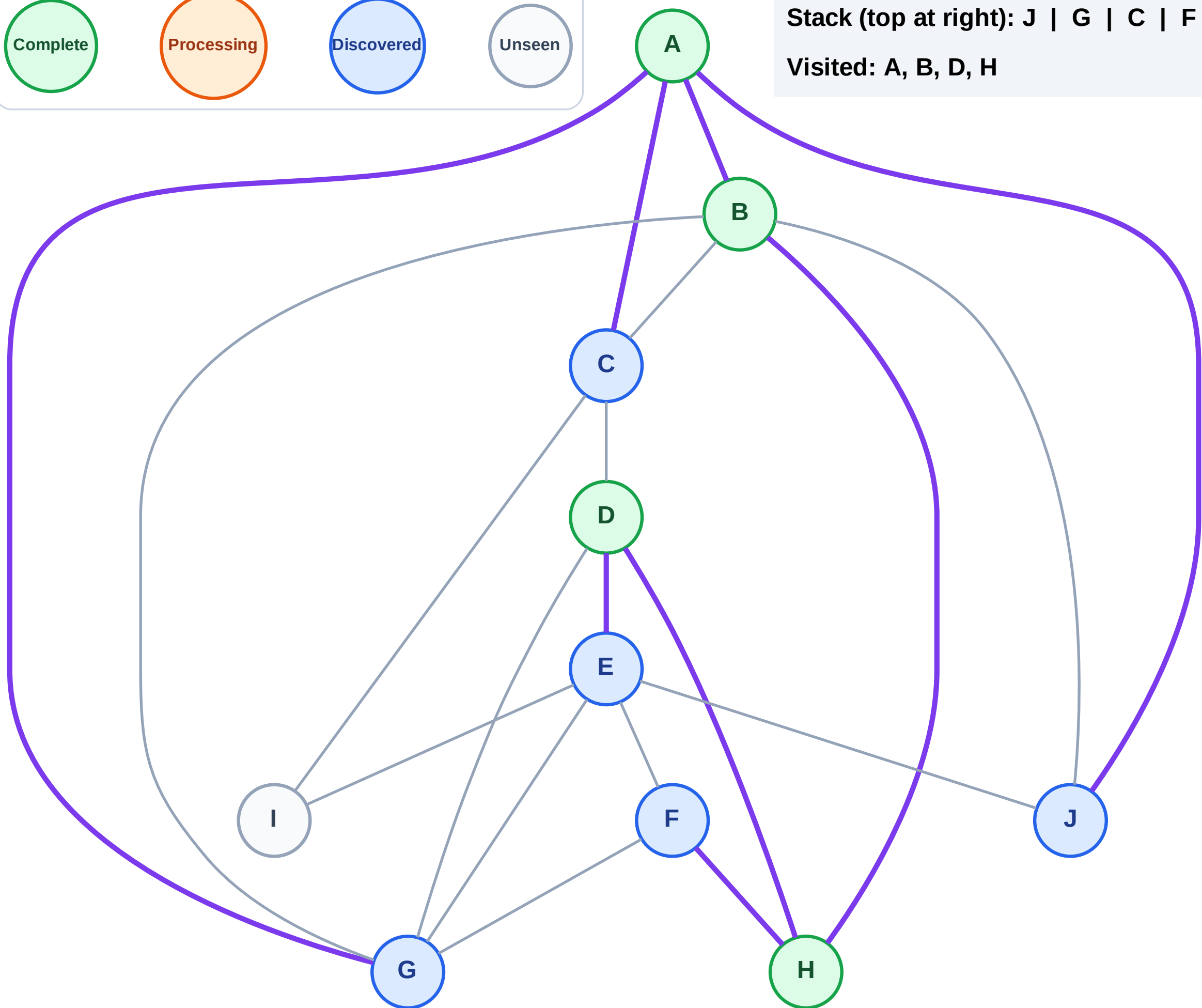
Step 9: Discover E from D

Legend



Stack (top at right): J | G | C | F | E

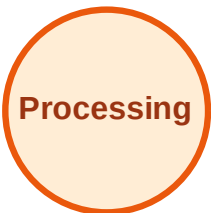
Visited: A, B, D, H



DFS Traversal

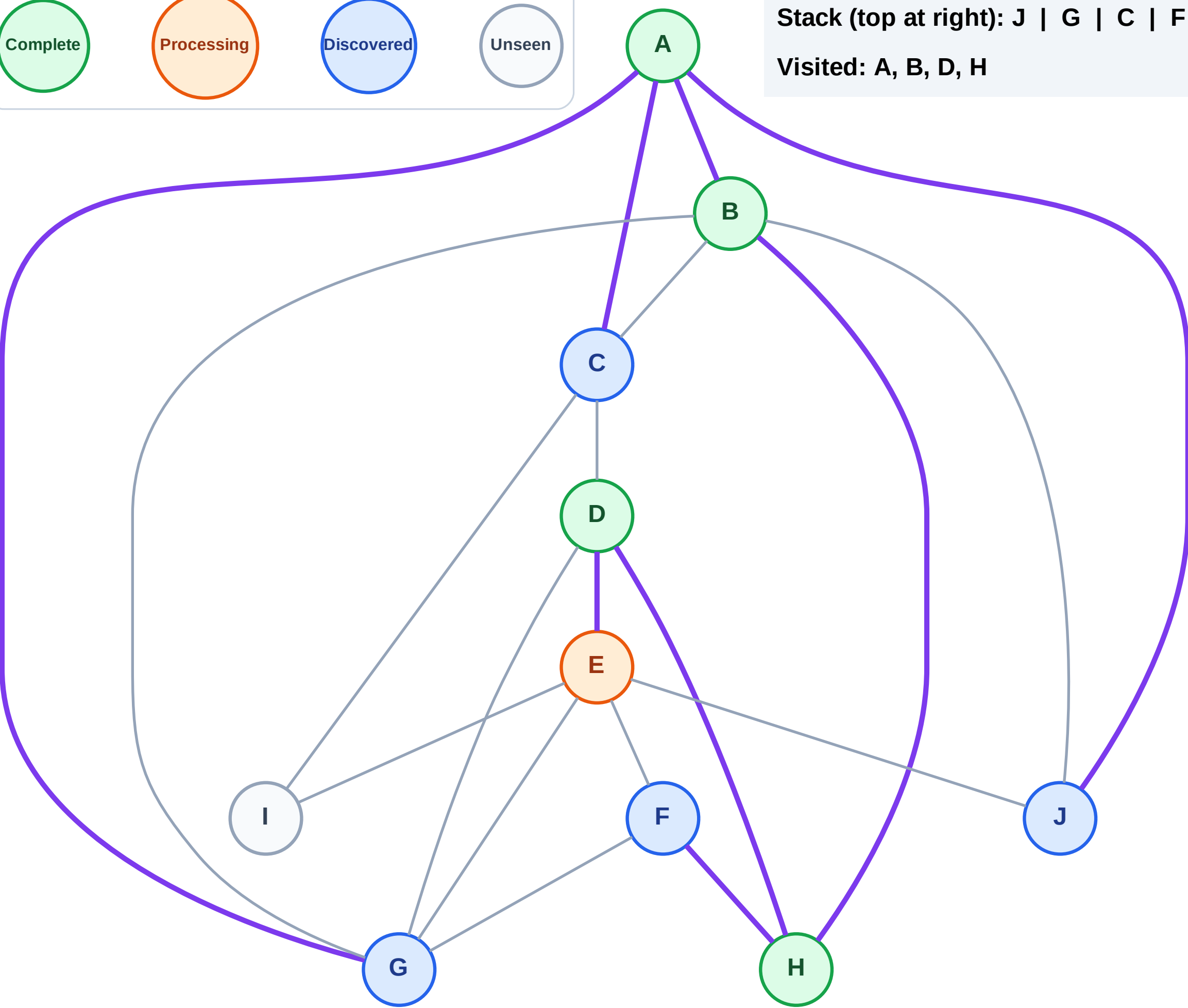
Step 10: Process node E

Legend



Stack (top at right): J | G | C | F

Visited: A, B, D, H



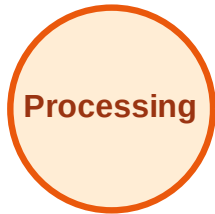
DFS Traversal

Step 11: Discover I from E

Legend



Complete



Processing



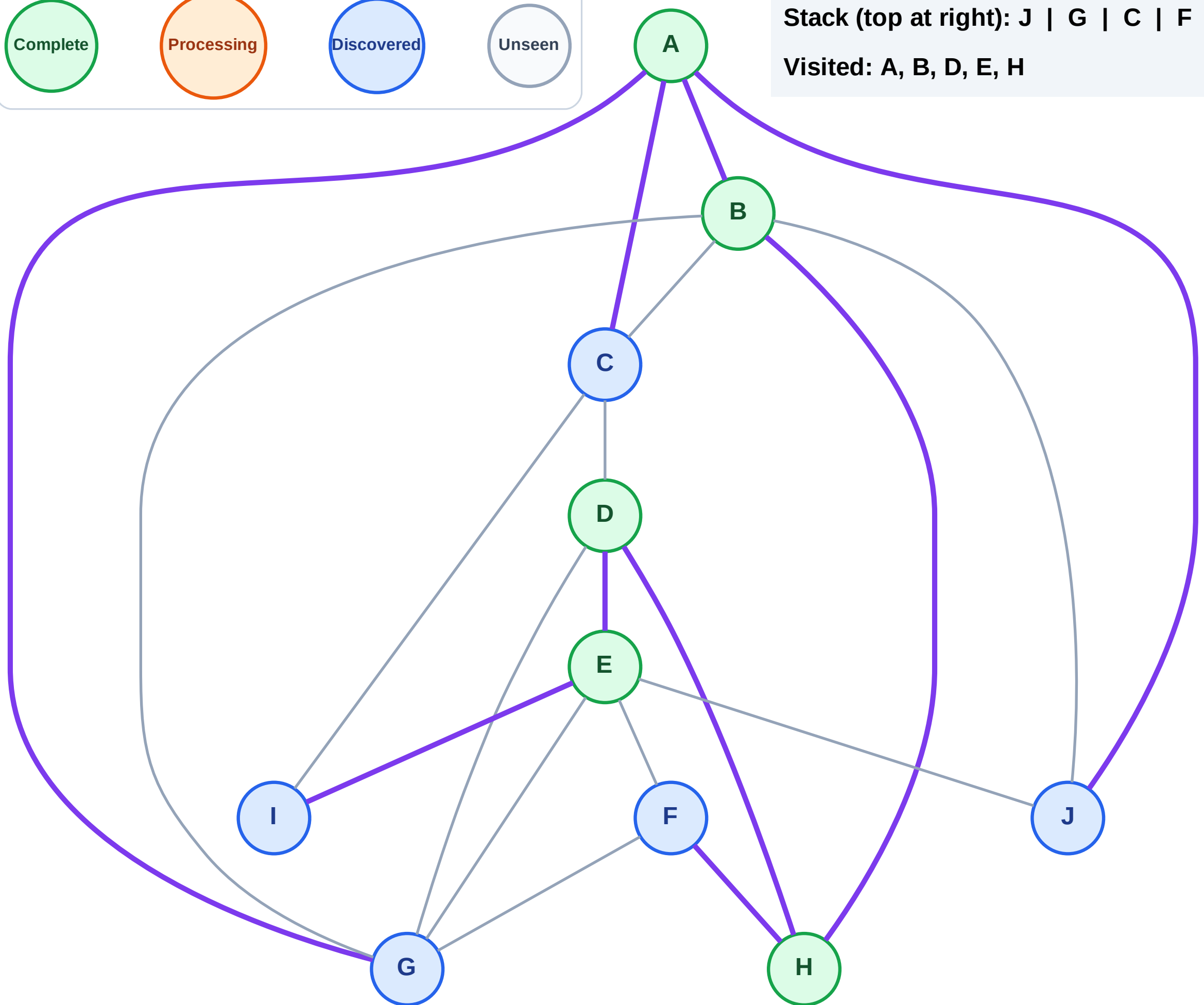
Discovered



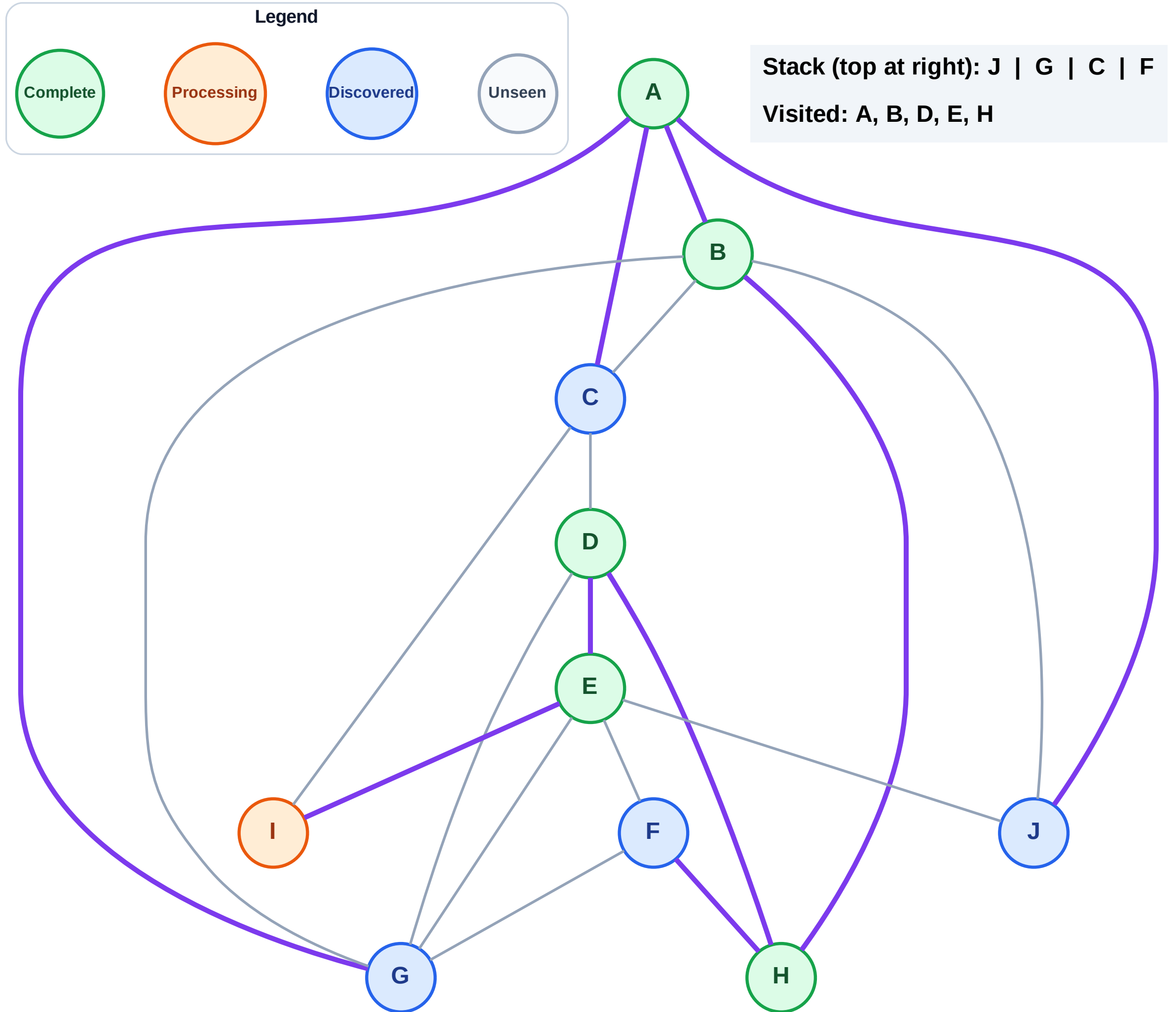
Unseen

Stack (top at right): J | G | C | F | I

Visited: A, B, D, E, H



Step 12: Process node I



DFS Traversal

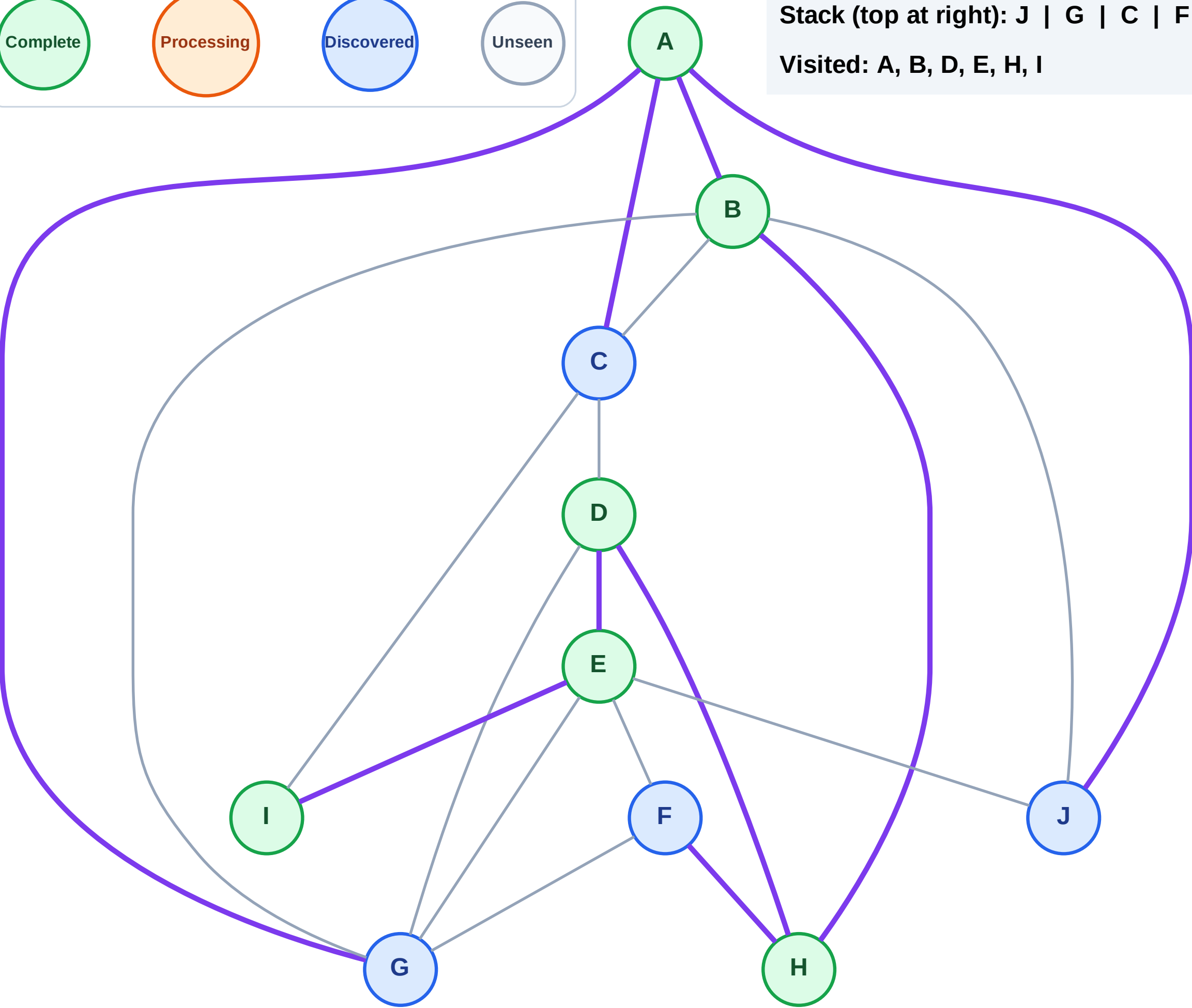
Step 13: No new neighbors from I

Legend



Stack (top at right): J | G | C | F

Visited: A, B, D, E, H, I



DFS Traversal

Step 14: Process node F

Legend



Complete



Processing



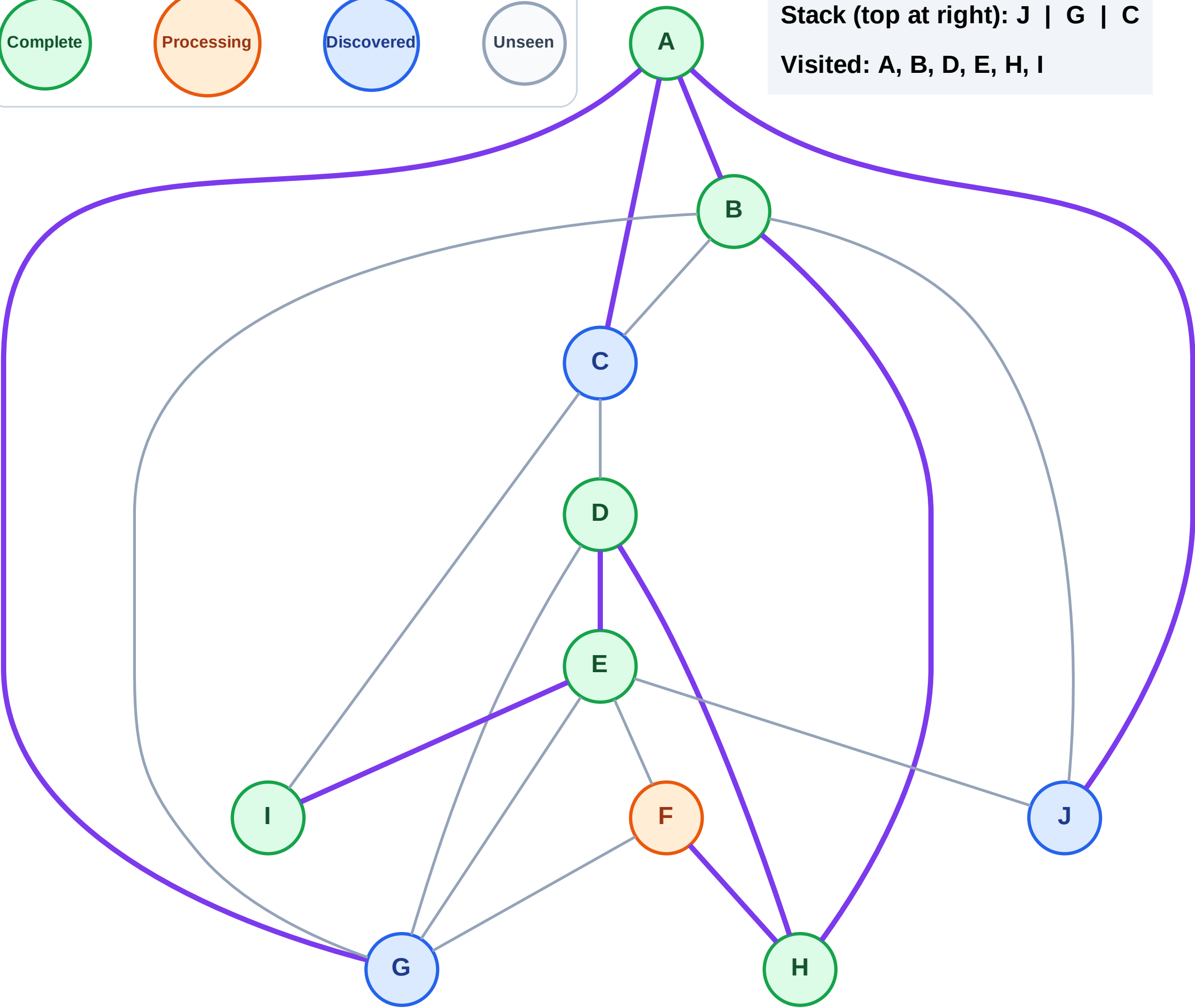
Discovered



Unseen

Stack (top at right): J | G | C

Visited: A, B, D, E, H, I



DFS Traversal

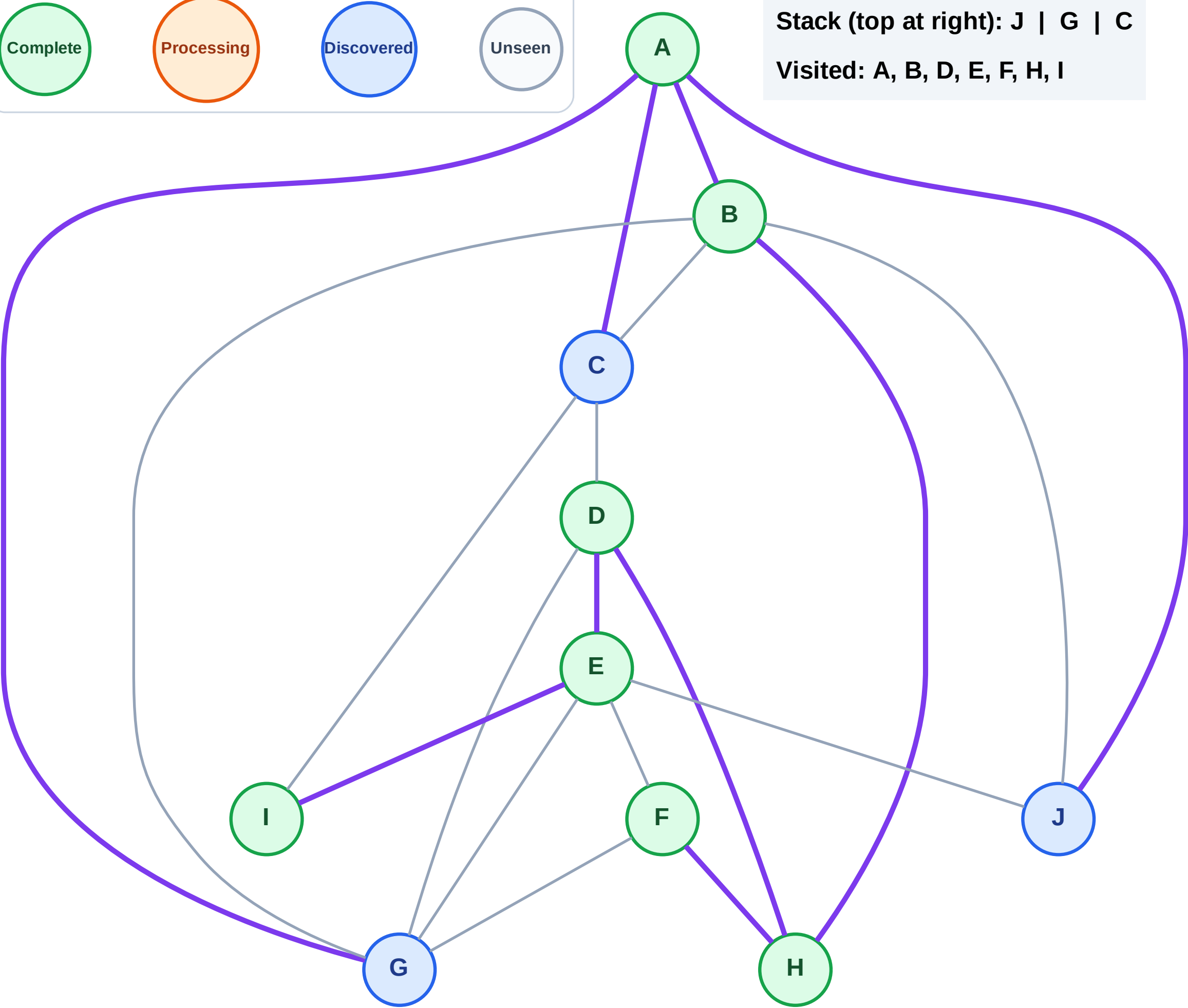
Step 15: No new neighbors from F

Legend



Stack (top at right): J | G | C

Visited: A, B, D, E, F, H, I



DFS Traversal

Step 16: Process node C

Legend



Complete



Processing



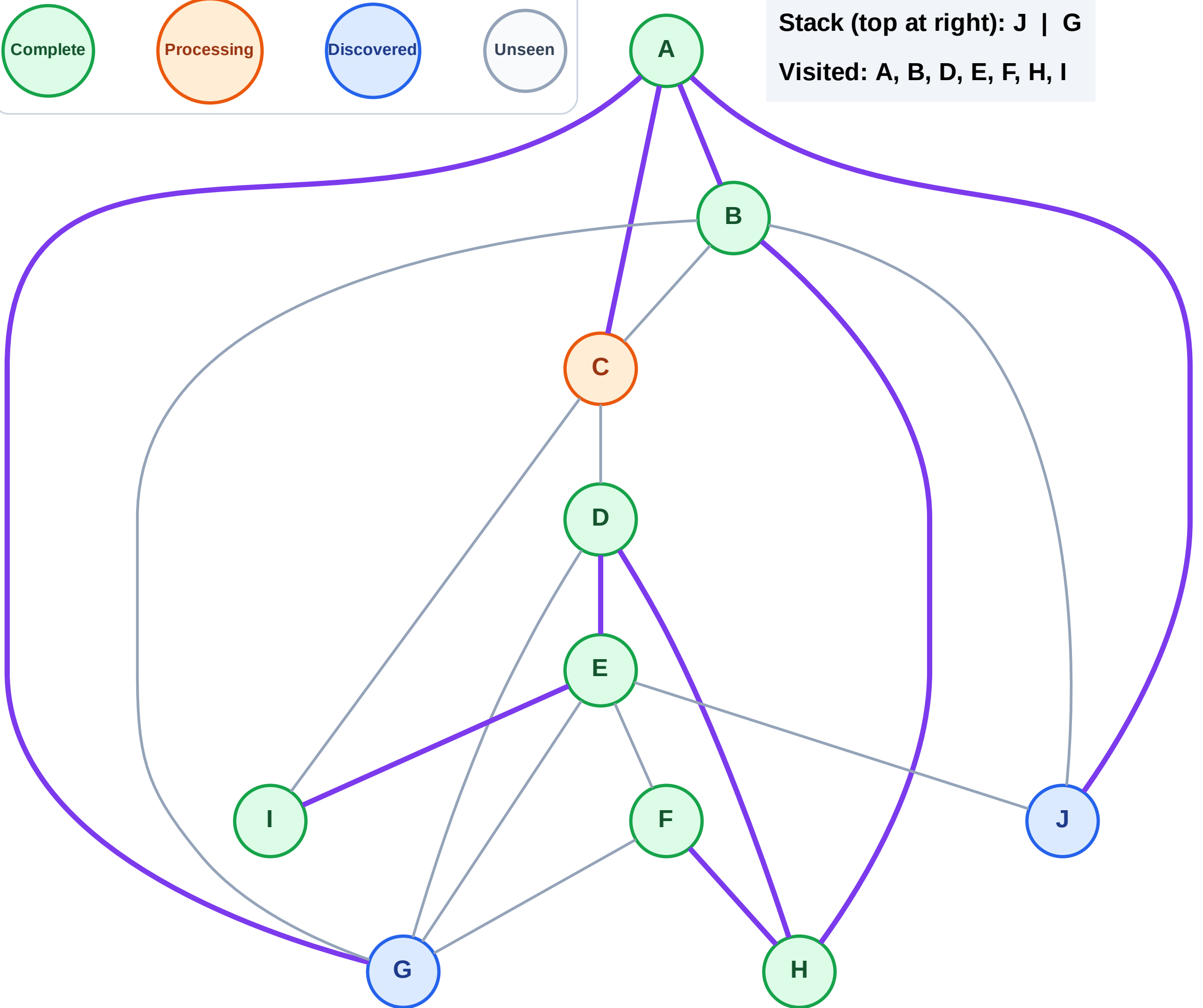
Discovered



Unseen

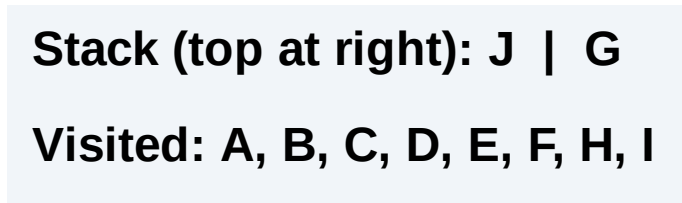
Stack (top at right): J | G

Visited: A, B, D, E, F, H, I



Step 17: No new neighbors from C

Step 17: No new neighbors from C



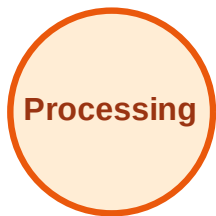
DFS Traversal

Step 18: Process node G

Legend



Complete



Processing



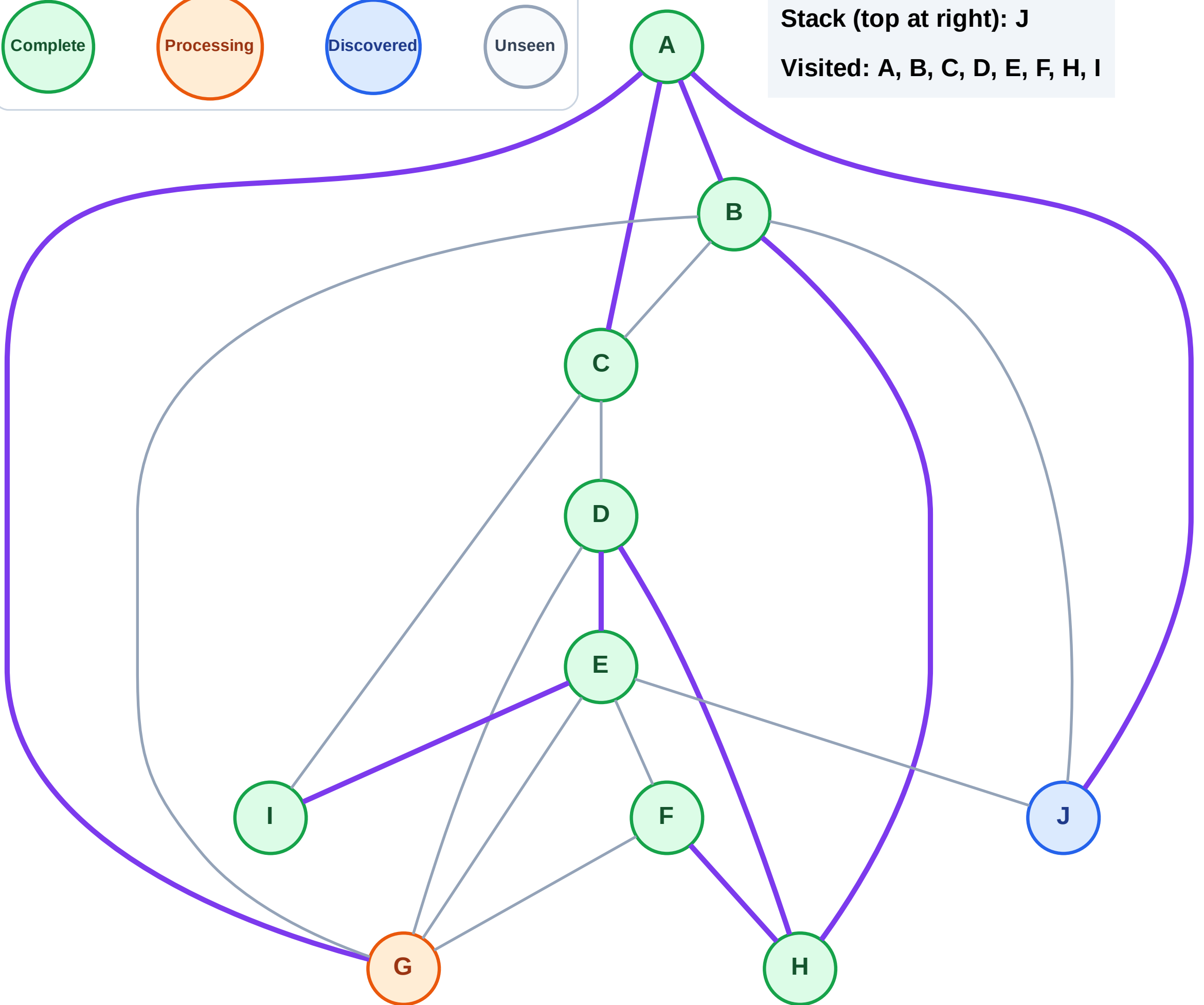
Discovered



Unseen

Stack (top at right): J

Visited: A, B, C, D, E, F, H, I



DFS Traversal

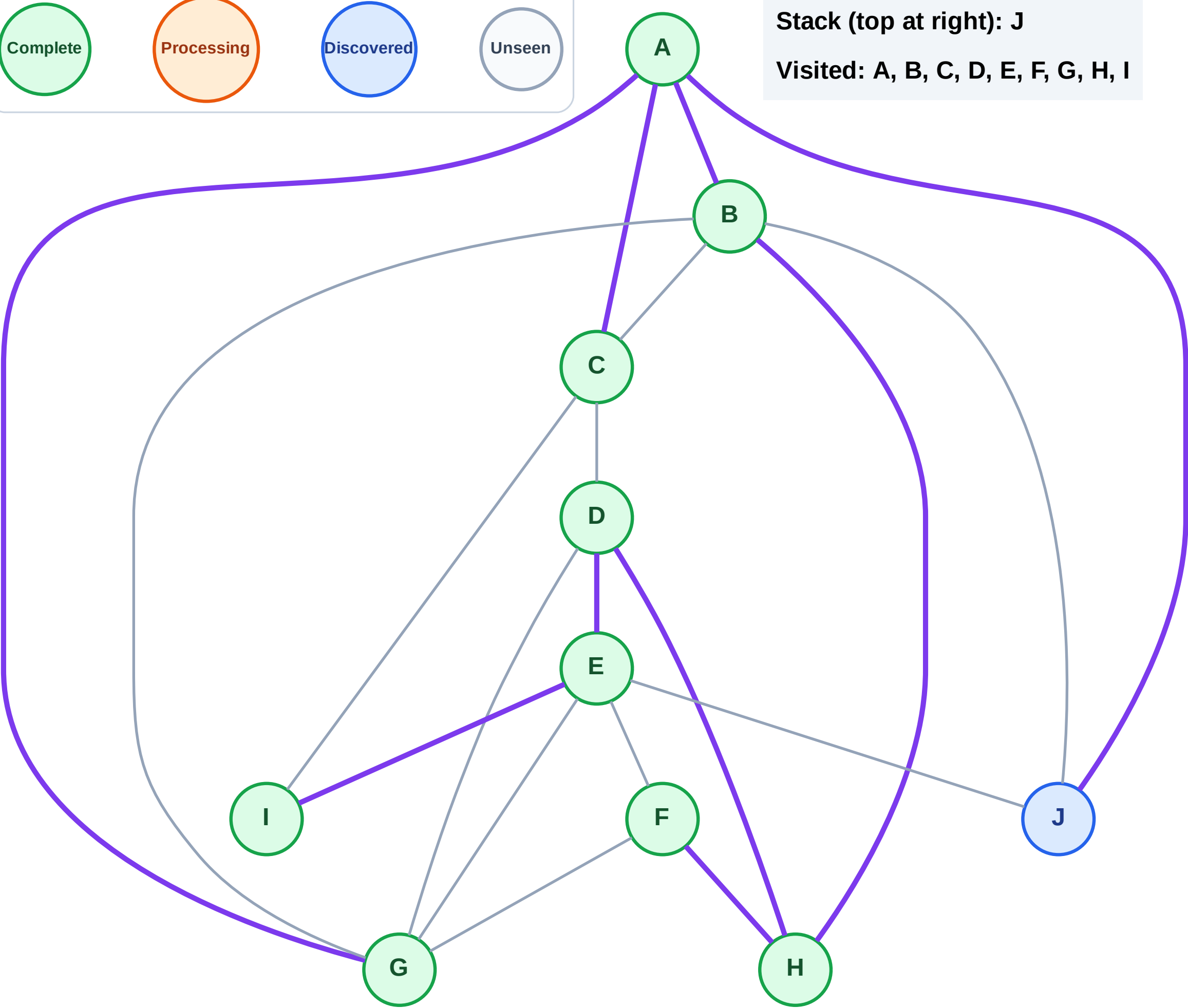
Step 19: No new neighbors from G

Legend



Stack (top at right): J

Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 20: Process node J

Legend



Complete



Processing



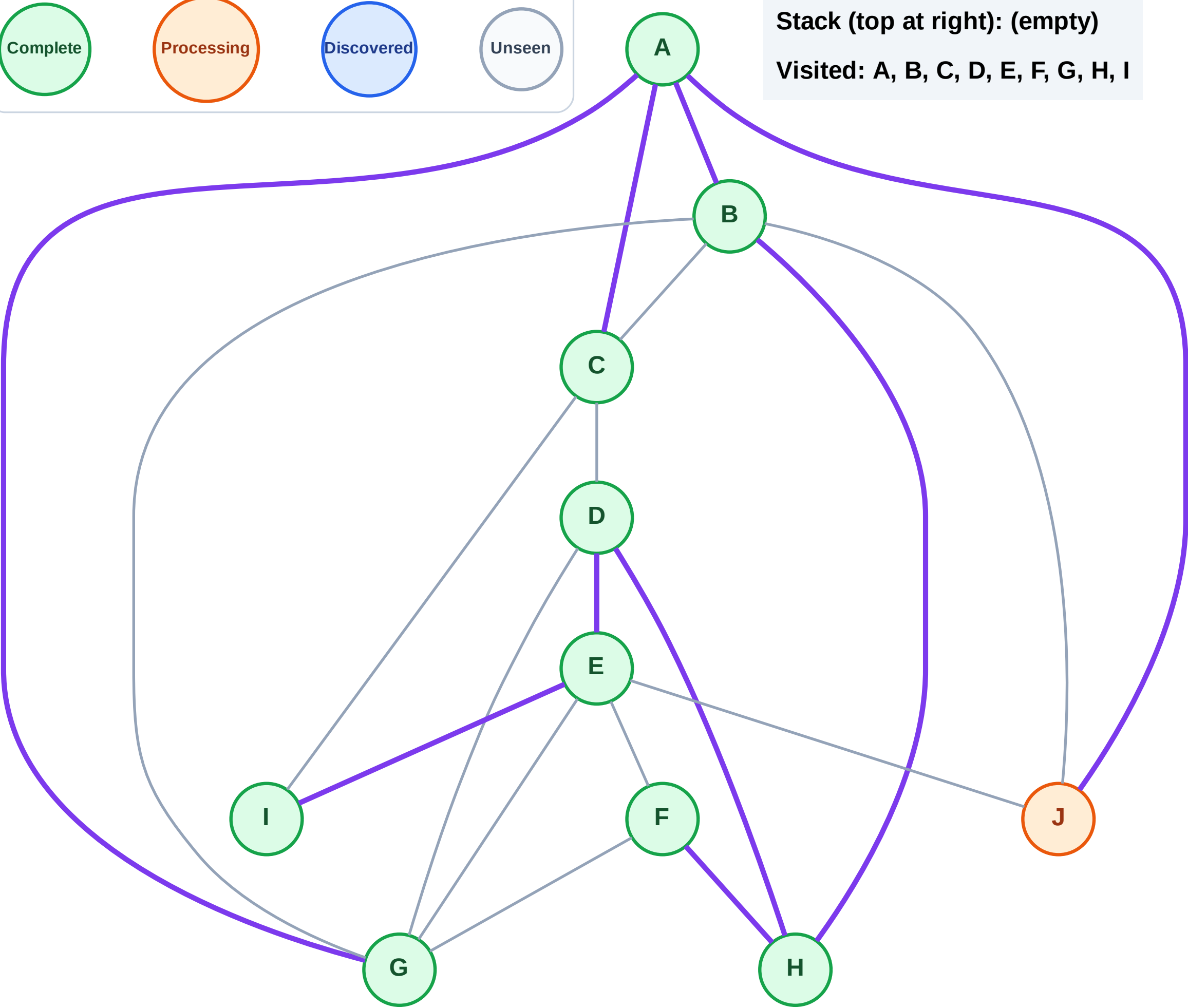
Discovered



Unseen

Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

